

Encoding and Calibration of Exact Norm Penalties in Mixed-Integer Convex Optimization

Abstract

Exact penalty methods move equality constraints that link the parts of a mixed-integer model into the objective as a norm of their violation. A finite coefficient that closes the duality gap is known to exist for mixed-integer linear programs and, under compactness and a Slater-type condition on the integer slices, for mixed-integer convex problems. But existence does not say how large the coefficient must be or how hard it is to find. We study both questions for coefficients written in ordinary binary encoding. First, the polynomial bit length known for linear constraints is lost once convex quadratic constraints are kept in the subproblem: a family with one binary variable, $d + 1$ continuous variables and $O(d \log(d + 1))$ input bits has least penalty $2^{2^d - 1}$, even with an optimal multiplier and although every integer slice has points at which all inequalities hold with slack $1/8$. Second, this growth needs both many continuous variables and many constraints that are nonlinear in them. For quadratic data that are convex in the continuous variables, affine linking equalities, finite bounds and a Slater condition on every integer slice that can satisfy the linking equalities, some integer coefficient with $(N + 1)2^{O(n)}$ bits is exact at multiplier zero for the ∞ - and 1-norms and preserves the set of minimizers, where N is the input length and n the number of continuous variables. This bound is tight up to the constant in the exponent, and $N^{O(k+1)}$ bits suffice when at most k inequalities are nonlinear in the continuous variables. Third, unless $P = NP$, no polynomial-time algorithm approximates the least penalty within any polynomial factor, even on a binary box with one equality, a known unique optimizer and a known sufficient coefficient. Finally, for finitely many compact convex slices with continuous convex objectives and affine residual maps, if the right-hand side is sampled from a rational grid with at least $4mK/\epsilon$ points per coordinate in a cube of radius σ , then with probability at least $1 - \epsilon$ either the perturbed problem is infeasible or the ∞ -norm coefficient $4mKM/(\sigma\epsilon)$ is exact for it at multiplier zero. Here m counts the linking equalities, K bounds the number of integer slices and M is a supplied bound on the objective range; the order K/ϵ is necessary.

1 Introduction

Many mixed-integer models keep most of their constraints in a set X and couple blocks, stages or scenarios through equalities $r(x) = 0$. Examples are linking constraints between subsystems, coupling constraints in market models, copy constraints between a stage and its state, and nonanticipativity constraints. We call the constraints that define X the *native constraints* and the equalities $r(x) = 0$ the *relaxed* (or *linking*) *equalities*. The vector x collects the integer and continuous variables. The problem is

$$v = \min\{f(x) : x \in X, r(x) = 0\},$$

and moving the relaxed equalities into the objective with a multiplier $\lambda \in \mathbb{R}^m$ and a norm penalty with coefficient $\rho \geq 0$ gives

$$L_\rho(\lambda) = \min_{x \in X} \{f(x) + \lambda^T r(x) + \rho \|r(x)\|\}, \quad D_\rho = \sup_{\lambda \in \mathbb{R}^m} L_\rho(\lambda).$$

Section 2 states the setting precisely (compact X , continuous f and r , a fixed norm). Weak duality gives $L_\rho(\lambda) \leq D_\rho \leq v$. For $\rho = 0$, D_0 is the ordinary Lagrangian dual, which for mixed-integer problems can leave a gap $D_0 < v$. The norm term, which gives the sharp augmented Lagrangian (not the squared term of the classical augmented Lagrangian), can close this gap with a finite coefficient. Such a coefficient is an *exact penalty*.

Exact penalty functions were introduced by Eremin [16] and Zangwill [36], and Burke [12] unified their theory. Rockafellar [32] showed that augmented Lagrangians remove duality gaps in nonconvex programming. Rockafellar and Wets [33] and Huang and Yang [25] developed the connection between augmented Lagrangian duality and exact penalization. For mixed-integer problems, finite exact penalties are known for bounded pure integer programs, rational MILPs, convex MIQPs, convex objectives over polyhedral mixed-integer sets, and mixed-integer nonlinear problems under regularity conditions on the integer slices [8, 10, 17, 22, 29]; Section 1.2 states these results precisely.

We use the following terms; Definition 2.1 states them formally. A coefficient ρ is *value-exact* if $D_\rho = v$, that is, if the penalized dual with the best multiplier has the optimal value. It is *exact at λ* if $L_\rho(\lambda) = v$ for that fixed multiplier. It is *minimizer-exact at λ* if the minimizers of $f + \lambda^T r + \rho \|r\|$ over X are exactly the optimal solutions of the original problem. Minimizer-exactness at λ implies exactness at λ , which implies value-exactness; Section 2 shows that the converses fail. The *least penalty* $\rho_* = \inf\{\rho \geq 0 : D_\rho = v\}$ is, when finite, the smallest value-exact coefficient, with the multiplier optimized. The *zero-multiplier threshold* $\rho_*(0)$ is, when finite, the smallest coefficient that is exact at $\lambda = 0$. We call a coefficient *sufficient* if it is value-exact, or exact at $\lambda = 0$ when the zero-multiplier threshold is meant.

Why the size of the coefficient matters. Exact norm penalties are used in decomposition methods, and each of them must choose the coefficient. In cutting-plane methods for multistage stochastic MILPs, augmented Lagrangian cuts with a norm as augmenting function give Lipschitz lower approximations of cost-to-go functions; they are tight for a suitable multiplier and large enough ρ , but arbitrarily large ρ may be needed near discontinuities of the value function [1]. Zhang and Sun [37] penalize copy constraints between a state and its local copy, assume that this penalty reformulation is exact (their Assumption 2), and their iteration bounds depend on the penalty sizes. For Lipschitz regularization in stochastic dual dynamic programming, which penalizes such copy constraints by a norm, a recent survey calls the choice of a sufficiently large coefficient an open challenge in practice [19]. In ℓ_1 augmented Lagrangian and ADMM decompositions of two-block MILPs, the penalty that guarantees an ε -accurate solution grows like $1/\varepsilon$, and such a large penalty slows the method [34]; in bundle methods for generalized augmented Lagrangian duals, adjusting the penalty parameter was the delicate part [13]; and penalty alternating direction heuristics increase weighted ℓ_1 penalties on copy constraints in an outer loop [20]. In ℓ_1 augmented Lagrangian pricing of nonconvex electricity markets, the coefficient is part of the price signal, and Wang et al. [35] seek a valid coefficient as close to the least one as is computationally feasible. Large coefficients also cost precision: floating-point MIP solvers suffer from wide ranges of data magnitudes [27], and quantum annealers rescale QUBO coefficients to a fixed range, so a large penalty compresses the gaps between objective values [28].

These uses raise two questions about a sufficient coefficient. We measure coefficients in ordinary binary encoding: a rational number is given by the binary representations of its numerator and its positive denominator.

- *Representation.* How many bits does a sufficient coefficient need, as a function of the encoding length of the instance?

- *Calibration.* How closely can the least penalty ρ_* be approximated, from above, by an efficient algorithm?

Bit length and numerical size are different measures: a coefficient with polynomially many bits may still be exponentially large as a number. A coefficient whose bit length is superpolynomial cannot even be written down in polynomial time.

Both questions are posed in the conclusion of Lefebvre and Schmidt [29]. They note that an exact penalty of polynomial size is known only for MIQPs and ask whether this extends to classes such as quadratically constrained quadratic problems. They also observe that the coefficient computed by their polynomial-time method is “too large to be useful in practical applications”, and they ask whether the smallest penalty that closes the duality gap can be computed in polynomial time, conjecturing that it cannot. We answer the first question negatively for convex quadratic native constraints when both the continuous dimension and the number of nonlinear constraints grow, and positively when either is fixed, under explicit variable bounds and a Slater condition on the integer slices (Assumption 4.1); for nonconvex quadratic constraints the question remains open (Section 8). We confirm the conjecture unless $P = NP$, even for approximation within any polynomial factor.

1.1 Main results

Table 1 summarizes the results. Throughout, N is the encoding length of an instance, n the number of continuous variables and m the number of relaxed equalities.

A doubly exponential lower bound. Theorem 3.1 gives, for each chain length d , an instance with one binary variable, $n = d + 1$ continuous variables, a linear objective, one relaxed equality, and convex quadratic and affine native constraints with coefficients from a fixed finite set; its sparse encoding length is $O(d \log(d + 1))$. Its least penalty is $2^{2^d - 1}$, even after optimizing the multiplier, so every exact rational coefficient, in any of the three senses, needs at least 2^d numerator bits. This holds although every integer slice has points with slack at least $1/8$, and a fixed additive accuracy below the objective range does not help (Corollary 3.3).

The mechanism is simple. A chain of convex quadratic inequalities $a_{i+1} \geq a_i^2$ lets the improving integer assignment come within $\delta_d = 2^{-2^d}$ of satisfying the relaxed equality, but never reach it. The other assignment has residuals of both signs at zero cost. A convex combination of its point with residual -1 and the improving point, weighted so that the residuals average to zero (a *balanced mixture*, Section 2), cancels every multiplier, and its average penalized cost is negative unless $\rho \geq 1/(2\delta_d)$. The numbers grow fast. For $d = 5$, with six continuous variables, the improving assignment violates the relaxed equality by $\delta_5 = 2^{-32} \approx 2.3 \cdot 10^{-10}$, below common feasibility tolerances, and the least penalty is 2^{31} . For $d = 10$ it is 2^{1023} , the largest power of two in IEEE double precision; for $d = 11$ it exceeds the double-precision range. Thus finite exactness with convex quadratic native constraints does not come with the polynomial encoding guarantee known for linear native constraints. The obstruction lies in the representation of the penalty, not in the original problem, whose optimal value is immediate.

Encoding upper bounds. For the positive results we fix the integer variables and look at the resulting *slices* of the native set. A slice is *equality-feasible* if it contains a point satisfying the relaxed equalities. The *refined Slater condition* requires, for every equality-feasible slice, a point of the slice that satisfies the relaxed equalities and strictly satisfies every native inequality that is nonlinear in the continuous variables; inequalities affine in the continuous variables may hold with equality, and nothing is assumed on the other slices. Assumption 4.1 asks for degree-two data

Result	Setting and hypotheses	Conclusion	Reference
Lower bound	One binary and $n = d + 1$ continuous variables; linear objective; $d - 1$ convex quadratic native inequalities ($d \geq 2$); coefficients from a fixed finite set; sparse encoding length $O(d \log(d + 1))$; points with slack $1/8$ on every integer slice	$\rho_* = 2^{2^d - 1}$, $\rho_*(0) = 2^{2^d}$; every value-exact rational coefficient has at least 2^d numerator bits; a gap at most $\epsilon < 1$ needs $2^d - O_\epsilon(1)$ bits	Thm. 3.1, Cor. 3.3
Lower bound in N	Same chain with seed $a_1 \geq 2^{-t}$, t of order N	least penalty with $\Omega(N2^n)$ bits	Rem. 3.5
Upper bound in n	Assumption 4.1: degree-two data, convex in the continuous variables for each integer assignment; finite bounds; nonempty feasible set; refined Slater condition; affine relaxed equalities	an integer ρ with $(N + 1)2^{O(n)}$ bits, minimizer-exact at $\lambda = 0$ for both $\ \cdot\ _\infty$ and $\ \cdot\ _1$	Thm. 4.2
Upper bound in k	Assumption 4.1 with at most k native inequalities nonlinear in the continuous variables; any number of affine rows	the same with $N^{O(k+1)}$ bits; for fixed n or fixed k such a coefficient can be printed in polynomial time	Thm. 5.1, Cor. 5.4
Calibration	Binary box, one equality, linear objective; known unique optimizer; $\rho = 1$ is exact at $\lambda = 0$	unless $P = NP$, no polynomial-time algorithm finds $\hat{\rho}$ with $\rho_* \leq \hat{\rho} \leq p(N)\rho_*$, and likewise for $\rho_*(0)$; testing value-exactness of $\rho = 1/3$ is coNP-complete	Thm. 6.4, Thm. 6.3
Strong hardness	Stable-set native set; coefficients in $\{-1, 0, 1\}$	$\rho_* = \alpha(G)$, so computing ρ_* is strongly NP-hard	Prop. 6.10
Perturbation	At most K compact convex slices; continuous convex objectives with supplied range M ; affine residuals; right-hand side sampled from a rational grid with at least $\lceil 4mK/\epsilon \rceil$ points per coordinate in a cube of radius σ	with probability at least $1 - \epsilon$, the sampled problem is infeasible or $4mKM/(\sigma\epsilon)$ is exact at $\lambda = 0$ for $\ \cdot\ _\infty$; the order K/ϵ is necessary	Thm. 7.3, Sec. 7.6

Table 1: Informal summary of the main results. ρ_* : least penalty (multiplier optimized); $\rho_*(0)$: zero-multiplier threshold; p : any fixed polynomial.

that are convex in the continuous variables for each integer assignment, explicit finite bounds on all variables, affine relaxed equalities, a nonempty feasible set and the refined Slater condition. Under this assumption, Theorem 4.2 gives an integer coefficient with $(N + 1)2^{O(n)}$ bits, and Theorem 5.1 gives one with $N^{O(k+1)}$ bits when at most k native inequalities are nonlinear in the continuous variables; the number of affine rows is arbitrary. In both cases one coefficient is minimizer-exact at multiplier zero for the infinity norm and the one-norm. By Remark 3.5, the worst case of the first bound is $N2^{\Theta(n)}$ bits.

Both proofs use Lemma 2.6, which reduces exactness at multiplier zero to a multiplier bound on each equality-feasible slice and a lower bound on the residual of each equality-infeasible slice; effective bounds from real algebraic geometry control both uniformly over all integer assignments.

Consequently, within the class of Assumption 4.1, coefficients with superpolynomially many bits require both a growing continuous dimension and a growing number of nonlinear native inequalities; the lower-bound instance has both. For fixed n , and separately for fixed k , a conservative coefficient can be printed in polynomial time (Corollary 5.4). These are existence bounds: the coefficient may be numerically huge, the constants are not evaluated, and nothing is said about solving the penalized subproblem.

Hardness of calibration. Theorem 6.4 considers pure binary linear programs with one equality: the native set is a binary box and the objective is linear. On these instances the unique optimal solution is known and $\rho = 1$ is exact at multiplier zero. Even so, unless $P = NP$, no polynomial-time algorithm can output a sufficient coefficient within any polynomial factor of the least penalty, or of the zero-multiplier threshold; Section 6 extends this to factors $2^{N^{1-\epsilon}}$ for every fixed $\epsilon \in (0, 1)$. Deciding whether the coefficient $\rho = 1/3$ is value-exact is coNP-complete (Theorem 6.3). The instances encode SUBSET SUM: when the target sum is attainable, some integer assignment that lowers the objective has a residual of magnitude 1; otherwise every such assignment has a residual of magnitude at least $\kappa - 1$, for a scaling integer κ that we take exponentially large. The hardness is weak, since a pseudo-polynomial dynamic program computes ρ_* ; a stable-set variant with coefficients in $\{-1, 0, 1\}$ makes exact computation strongly NP-hard (Proposition 6.10). The message is that a safe coefficient can be easy to find while a nearly least one is hard to certify.

Right-hand-side perturbations. Theorem 7.3 considers at most K compact convex slices with continuous convex objectives of range M and affine residual maps. If the right-hand side is sampled from a rational grid with at least $\lceil 4mK/\epsilon \rceil$ points per coordinate in a cube of radius σ , then with probability at least $1 - \epsilon$ the sampled problem is infeasible or the infinity-norm coefficient $4mKM/(\sigma\epsilon)$ is exact at multiplier zero. If the number of grid points per coordinate is the smallest power of two that is at least $\lceil 4mK/\epsilon \rceil$, the grid points and the coefficient have encoding length polynomial in the supplied rational data, in m and in $\log K$ (Remark 7.4). The mechanism is that large penalties arise only when the right-hand side lies near the boundary of the set of residuals attainable on some slice; a random right-hand side avoids these boundaries with high probability, and a convex repair argument turns the distance into a coefficient. The guarantee concerns the perturbed problem, whose optimal integer assignment may differ from the original one. The coefficient is numerically of order K , which can be exponential in the number of integer variables, and Section 7.6 shows that the order K/ϵ is necessary and that the expected least penalty can be infinite.

1.2 Related work

Exact penalties and augmented Lagrangian duality. The dual of Rockafellar [32] maximizes jointly over the multipliers and the penalty parameter; we fix ρ , optimize λ , and ask for the least ρ at which the optimized value is exact.¹ Section 2 relates the threshold $\rho_*(\lambda)$ to the exact penalty representations of the sharp Lagrangian in Rockafellar and Wets [33, Chapter 11]. In the study of global saddle points of augmented Lagrangians for continuous cone-constrained problems, Dolgopoliik [14] uses the term “least exact penalty parameter”; our least penalty ρ_* is the analogous quantity for mixed-integer problems, taken after optimizing the multiplier.

Exactness and encoding for mixed-integer problems. Boland and Eberhard [10] prove finite-penalty exactness for bounded pure integer programs under their assumptions on the augmenting

¹Theorem, section and example numbers of preprints and author manuscripts refer to the versions identified in the references.

function, and Feizollahi et al. [17] prove norm-penalty exactness for rational MILPs. Gu et al. [22, Theorem 11] show that for a convex quadratic objective over a rational mixed-integer polyhedron some exact norm penalty has polynomial encoding length. Bhardwaj et al. [8, Theorems 3.1–3.3] give value-function proofs of existence with data-dependent penalty parameters for rational MILPs and for MIQPs with bounded integer variables. For convex objectives over polyhedral mixed-integer sets they prove exactness when the integer variables are bounded on the feasible set or when the objective and the continuous relaxation share no nonzero recession direction, and they give an explicit bound under strong convexity and smoothness. Lefebvre and Schmidt [29, Theorem 14] prove that a finite penalty is exact at each fixed multiplier under compactness, a differentiable convex objective, convex slices and a Slater condition on every equality-feasible slice; the instance of Theorem 3.1 satisfies these assumptions. Their Theorem 15 computes, in polynomial time, an infinity-norm penalty that is exact at a given rational multiplier for MILPs with integer data, and their Table 1 remarks that the extension to convex MIQPs is easy. For native constraints linear in all variables and a jointly convex quadratic objective, polynomial encoding length of an exact penalty is therefore prior work [22, Theorem 11], and so is polynomial-time output of an exact coefficient for MILPs with integer data [29, Theorem 15]. Appendix A discusses their Example 13: under their supremum-based Definition 1 it is value-exact at every penalty, but no multiplier attains the dual value.

Squaring chains and bit complexity. The squaring chain and its small residuals are classical; Bienstock et al. [9, Example 6.1] attribute it to Ramana and Khachiyan. Example 6.2 of the same paper, a nonconvex QCQP with a point of doubly exponentially small violation, already implies large zero-multiplier norm penalties. Beck et al. [7, Sections 2 and 4, Result 4] use a bounded convex squaring chain with Slater regularity to show that bilevel problems are sensitive to feasibility errors. To our knowledge, what is new in Theorem 3.1 is the exact threshold of the optimized dual for a compact convex-quadratic model with one binary variable and uniform strict points, together with its fixed-accuracy consequence. Residuals of opposite signs on the cost-free slice make the bound hold for every multiplier.

Real-algebraic tools. The upper bounds rest on established tools: radii of balls meeting the connected components of semialgebraic sets [5] and quantitative quantifier elimination [4, 6]. Algebraic bounds that depend on the number of quadratic constraints appear in Grigoriev and Pasechnik [21] and Kamminga and Rudolph [26]; Section 5 compares them with our argument. To our knowledge, what is new here are the resulting encoding bounds for exact penalties, including the dependence on the number of native inequalities that are nonlinear in the continuous variables while the number of affine rows is arbitrary.

Hardness of choosing penalties. Hardness of choosing penalties, even when the constrained optimum is known, predates this paper. Alessandrini et al. [2, Section IV.A, Lemma 1] prove hardness of choosing penalties for exact QUBO reformulations. Their reduction imposes $x = 0$ with a squared residual and a quadratic objective; on binary variables their penalty $M \sum_i x_i$ is reproduced by a linear multiplier with augmentation coefficient zero, so it does not give a positive optimized threshold. Mirkarimi et al. [30, Section 2.2] note that a universally successful, polynomially tuned linear Ising penalty for knapsack would imply $P = NP$, allowing that no linear penalty may work on some instances. On the family of Theorem 6.4, $D_0 = -1 < v$, so no purely linear (Lagrangian) penalty is ever exact and the norm term is essential. Alessandrini et al. [3, Definition 1 and Theorem 2] study probabilistic feasibility and objective guarantees when exact Gibbs sampling

is used for approximate optimization. Safe conservative penalties are also known: Gusmeroli and Wiegele [24, Sections 5–6] obtain exact squared penalties for binary quadratic problems with integer-data linear equality constraints from objective bounds and residual separation ($\|Ax - b\|^2 \geq 1$ by integrality). To our knowledge, what is new in Theorem 6.4 is the polynomial-factor conclusion for a norm-penalty threshold that is always positive and is taken after unrestricted optimization of the multiplier, under the restrictions above: a binary box, one equality, a linear objective, a known unique optimizer and a known sufficient coefficient.

Conditioning and perturbation. Distance to the boundaries of convex sets is a classical conditioning device [15]; Bürgisser and Amelunxen [11, Theorem 3.3 and Corollary 3.4] bound the relative volume of boundary neighbourhoods of spherically convex sets within spherical caps. Theorem 7.3 is an explicit application to exact penalties, with a rational grid, an encoding bound and the infeasibility alternative; its Gaussian variant (Proposition 7.10) has the infeasibility alternative but no grid or encoding statement. Their ingredients are known: the boundary-avoidance principle, the convex repair argument and, for the Gaussian variant, the tube estimate [15, Lemma 2.3.4]. The relation between convex exact penalties and multiplier bounds is classical [18, Theorem 4.2], and Lipschitz value bounds across discrete choices appear in mixed-integer optimal control [23, Theorem 3].

Organization. Section 2 defines the exactness notions and the two thresholds and proves the balanced-mixture, one-equality and slice results. Section 3 proves the lower bound and discusses which features of the formulation it depends on. Sections 4 and 5 prove the encoding upper bounds in the continuous dimension and in the number of nonlinear native inequalities. Section 6 proves the calibration hardness, and Section 7 the perturbation results. Section 8 discusses consequences, scope and open questions. Appendix A treats Example 13 of Lefebvre and Schmidt [29], and Appendix B gives a symmetric two-binary variant of the lower bound in which the optimized and zero-multiplier thresholds coincide.

2 Exactness notions and residual geometry

This section fixes the setting and the vocabulary of the paper: three notions of exactness and two thresholds. It then proves three tools. Balanced mixtures describe the optimized dual value in primal terms and supply the lower-bound witnesses of Sections 3, 6 and 7; one relaxed equality admits explicit thresholds; and a slice-wise sufficient condition at multiplier zero underlies the upper bounds of Sections 4 and 5.

2.1 Setting

Let X be a nonempty compact subset of a finite-dimensional Euclidean space, and let $f : X \rightarrow \mathbb{R}$ and $r : X \rightarrow \mathbb{R}^m$ be continuous. The set X is the *native set*: it collects the *native constraints*, which are kept in the penalized subproblem. The equalities $r(x) = 0$ are the *relaxed equalities*; they are moved into the penalty. In applications they are linking or coupling constraints, so we also call them linking equalities. In Sections 4 and 5 they have the form $r(x, z) = Ax + Bz - b$. We assume that the feasible set $F = \{x \in X : r(x) = 0\}$ is nonempty. Fix a norm $\|\cdot\|$ on $\mathbb{R}^m$, with dual norm $\|\cdot\|_*$. For $\rho \geq 0$ and $\lambda \in \mathbb{R}^m$ let

$$v = \min_{x \in F} f(x), \quad L_\rho(\lambda) = \min_{x \in X} \{f(x) + \lambda^T r(x) + \rho \|r(x)\|\}, \quad D_\rho = \sup_{\lambda \in \mathbb{R}^m} L_\rho(\lambda). \quad (1)$$

Here v is the optimal value, $L_\rho(\lambda)$ is the penalized value at the multiplier λ , and D_ρ is the optimized (dual) value. The minima exist because F is closed in the compact set X and f and r are continuous. As a minimum of functions that are affine in λ , L_ρ is concave and finite on $\mathbb{R}^m$, hence continuous. On F the penalized objective equals f , so weak duality holds:

$$L_\rho(\lambda) \leq D_\rho \leq v \quad \text{for all } \rho \geq 0, \lambda \in \mathbb{R}^m.$$

Thus D_ρ is the supremum of the lower bounds on v that the multipliers give at the coefficient ρ .

Slices. In mixed-integer models some coordinates of the points of X are integer variables; Sections 4 and 5 write a point as (x, z) with continuous part x and integer part z . The *slice* X_z is the set of native points whose integer part equals z . When convenient we identify X_z with the set of continuous parts of its points. Because X is compact, only finitely many slices are nonempty, and each of them is compact. A slice is *equality-feasible* if it meets F , that is, if it contains a point with $r = 0$; otherwise it is *equality-infeasible*. A nonempty equality-infeasible slice has positive *residual separation* $\delta_z = \min_{x \in X_z} \|r(x)\|$, by compactness and continuity.

Encoding conventions. Statements about encoding length use ordinary binary encoding. A rational number is encoded by the binary representations of its integer numerator and its positive integer denominator. An instance is given by explicit rational coefficients, and N denotes its binary encoding length. A sparse encoding records the variable indices as well as the nonzero coefficients; Assumption 4.1 fixes the encoding of quadratic data in Sections 4 and 5. The norm and the relaxed equalities are fixed parts of the formulation; rescaling r or changing the norm gives a different instance, which can have different thresholds (Section 3.4).

2.2 Exactness notions and thresholds

Definition 2.1 (Exactness notions and thresholds). Let $\rho \geq 0$ and $\lambda \in \mathbb{R}^m$.

- ρ is *value-exact* if $D_\rho = v$.
- ρ is *exact at* λ if $L_\rho(\lambda) = v$.
- ρ is *minimizer-exact at* λ if the set of minimizers of $f + \lambda^T r + \rho \|r\|$ over X equals $\operatorname{argmin}_{x \in F} f(x)$.

The *least penalty* is $\rho_* = \inf\{\rho \geq 0 : D_\rho = v\}$. The *threshold at* λ is $\rho_*(\lambda) = \inf\{\rho \geq 0 : L_\rho(\lambda) = v\}$, and $\rho_*(0)$ is the *zero-multiplier threshold*. We use the convention $\inf \emptyset = +\infty$.

Minimizer-exactness is what primal recovery needs: every minimizer of the penalized subproblem is then an optimal solution of the original problem.

The threshold at a fixed multiplier. For every $\lambda \in \mathbb{R}^m$,

$$\rho_*(\lambda) = \max \left\{ 0, \sup_{x \in X \setminus F} \frac{v - f(x) - \lambda^T r(x)}{\|r(x)\|} \right\}, \quad (2)$$

where the supremum of the empty set is $-\infty$. Indeed, by weak duality, $L_\rho(\lambda) = v$ holds exactly when $f(x) + \lambda^T r(x) + \rho \|r(x)\| \geq v$ for every $x \in X$. On F this inequality holds because $f \geq v$ there. For $x \notin F$ we have $\|r(x)\| > 0$, and the inequality says that ρ is at least the ratio in (2). Two consequences follow.

- The coefficients exact at λ form the interval $[\rho_*(\lambda), \infty)$ when $\rho_*(\lambda)$ is finite. The threshold $\rho_*(\lambda)$ may be $+\infty$; then no coefficient is exact at λ (Example 2.5).
- Every $\rho > \rho_*(\lambda)$ is minimizer-exact at λ : at each $x \notin F$, the extra term $(\rho - \rho_*(\lambda))\|r(x)\|$ is positive, so the penalized objective exceeds v , while on F it equals f .

Implications between the notions. For every $\rho \geq 0$ and $\lambda \in \mathbb{R}^m$,

$$\text{minimizer-exact at } \lambda \implies \text{exact at } \lambda \implies \text{value-exact}.$$

For the first implication, a point of $\operatorname{argmin}_{x \in F} f(x)$ is then a penalized minimizer, with penalized value v . For the second, $v = L_\rho(\lambda) \leq D_\rho \leq v$; in this case λ attains the dual supremum. Consequently $\rho_* \leq \rho_*(\lambda)$ for every λ . Neither converse holds. Example 2.5 and Appendix A give value-exact coefficients that are exact at no multiplier, because the dual supremum is not attained. Exactness at λ does not imply minimizer-exactness at λ either: at $\rho = \rho_*(\lambda)$, infeasible points can attain the value v . Such ties occur in Sections 3 and 6.

Relation to the sharp Lagrangian. The function $L_\rho(\lambda)$ is the dual function of the sharp Lagrangian [33, Example 11.58]. For compact X their level-boundedness condition holds. Then λ supports an exact penalty representation in the sense of Rockafellar and Wets [33, Definition 11.60] exactly when $\rho_*(\lambda)$ is finite, and the adequate penalty thresholds are then the positive $\rho \geq \rho_*(\lambda)$ [33, Theorem 11.61]. For $\lambda = 0$ this is calmness from below of the value function, that is, of the optimal value as a function of the right-hand side of the relaxed equalities [33, Example 11.62(b)]. The least penalty ρ_* takes the multiplier into account, and it can be smaller than every $\rho_*(\lambda)$ (Example 2.5).

2.3 Balanced mixtures and the least penalty

A *balanced mixture* π is a finite family of native points $x_i \in X$ with weights $\theta_i \geq 0$ such that

$$\sum_i \theta_i = 1 \quad \text{and} \quad \sum_i \theta_i r(x_i) = 0.$$

Its *average cost* and *average residual norm* are

$$c_\pi = \sum_i \theta_i f(x_i), \quad s_\pi = \sum_i \theta_i \|r(x_i)\|.$$

Let $\mathcal{B}$ be the set of all balanced mixtures. The residuals cancel on average, so the multiplier term $\lambda^T r$ averages to zero over a balanced mixture. Hence, for every λ ,

$$L_\rho(\lambda) \leq \sum_i \theta_i [f(x_i) + \lambda^T r(x_i) + \rho \|r(x_i)\|] = c_\pi + \rho s_\pi.$$

A balanced mixture with $c_\pi + \rho s_\pi < v$ therefore certifies a duality gap that no multiplier can remove. Proposition 2.2 shows that the best such certificate gives exactly D_ρ . The lower-bound witnesses in Sections 3, 6 and 7 are balanced mixtures.

The convexification argument behind Proposition 2.2 is that of Boland and Eberhard [10] and Feizollahi et al. [17]; Feizollahi et al. [17, Theorem 1] use a lifted convex hull under a relative-interior condition. The residual-space version below, for compact X , needs no such condition and does not assume that the dual supremum is attained.

Proposition 2.2 (Balanced mixtures). *For every $\rho \geq 0$, the optimized value D_ρ is the minimum of $c_\pi + \rho s_\pi$ over all balanced mixtures π , and this minimum is attained by a balanced mixture of at most $m + 1$ points:*

$$D_\rho = \min \left\{ \sum_{i=1}^{m+1} \theta_i [f(x_i) + \rho \|r(x_i)\|] : \begin{array}{l} x_i \in X, \theta_i \geq 0, \\ \sum_i \theta_i = 1, \sum_i \theta_i r(x_i) = 0 \end{array} \right\}. \quad (3)$$

Proof. We show that D_ρ is the lowest point above the origin of a convex hull in $\mathbb{R}^m \times \mathbb{R}$, and then reduce the number of points.

Let $\mathcal{G} = \{(r(x), f(x) + \rho \|r(x)\|) : x \in X\} \subseteq \mathbb{R}^m \times \mathbb{R}$. It is compact as a continuous image of X , so its convex hull $\mathcal{C} = \text{conv } \mathcal{G}$ is compact. The points of $\mathcal{C}$ of the form $(0, t)$ are exactly the points $(0, c_\pi + \rho s_\pi)$ with $\pi \in \mathcal{B}$. Such points exist, since a point of F is a balanced mixture by itself. Let $w = \min\{t : (0, t) \in \mathcal{C}\}$. Then $c_\pi + \rho s_\pi \geq w$ for every $\pi \in \mathcal{B}$, with equality for some π .

First, $D_\rho \leq w$: averaging over a balanced mixture with value w cancels λ , so $L_\rho(\lambda) \leq w$ for every λ .

Second, $D_\rho \geq w$. Let $a < w$. Then $(0, a) \notin \mathcal{C}$, and strict separation of the point $(0, a)$ from the compact convex set $\mathcal{C}$ gives $(u, \beta) \in \mathbb{R}^m \times \mathbb{R}$ with

$$u^T \zeta + \beta t > \beta a \quad \text{for all } (\zeta, t) \in \mathcal{C}.$$

At $(\zeta, t) = (0, w)$ this reads $\beta w > \beta a$, so $\beta > 0$. At the points of $\mathcal{G}$, after division by β , it reads $f(x) + (u/\beta)^T r(x) + \rho \|r(x)\| > a$ for every $x \in X$. The minimum over the compact set X is attained, so $L_\rho(u/\beta) > a$ and $D_\rho > a$. Letting $a \uparrow w$ gives $D_\rho = w$. No attaining multiplier was used.

Finally, the support. By Carathéodory's theorem in $\mathbb{R}^{m+1}$, the point $(0, w)$ is a convex combination of at most $m + 2$ points of $\mathcal{G}$; take one with positive weights θ_i at points x_i . If there are $m + 2$ points, the vectors $(1, r(x_i)) \in \mathbb{R}^{m+1}$ are linearly dependent: some $\xi \neq 0$ satisfies $\sum_i \xi_i = 0$ and $\sum_i \xi_i r(x_i) = 0$. Because its entries sum to zero, ξ has a negative entry. For every small $|\nu|$, the weights $\theta + \nu \xi$ again form a balanced mixture, whose value $\sum_i (\theta_i + \nu \xi_i) [f(x_i) + \rho \|r(x_i)\|]$ is affine in ν , at least w , and equal to w at $\nu = 0$. Hence the value does not depend on ν . Increase ν from zero until the first weight with $\xi_i < 0$ vanishes. The value stays w and at most $m + 1$ points remain. Padding with zero weights gives the form (3). $\square$

The average in (3) contains the norms of the individual residuals, not the norm of their average, which is zero.

The support bound $m + 1$ is sharp. Let X consist of $m + 2$ points: one with $(r, f) = (0, 0)$, and $m + 1$ points with cost -1 and residuals $e_1, \dots, e_m$ and $-\sum_j e_j$, where e_j are the unit vectors. At $\rho = 0$ a balanced mixture must give equal weight to the last $m + 1$ points, so its value is -1 times their total weight. The optimal mixture gives weight $1/(m + 1)$ to each of them, with value -1 , while every balanced mixture of at most m points has value 0.

Corollary 2.3. *The least penalty is*

$$\rho_* = \max \left\{ 0, \sup_{\pi \in \mathcal{B}: s_\pi > 0} \frac{v - c_\pi}{s_\pi} \right\}, \quad (4)$$

where the supremum of the empty set is $-\infty$. The supremum may be restricted to balanced mixtures of at most $m + 1$ points. If ρ_* is finite, it is value-exact.

Proof. By Proposition 2.2, $D_\rho \geq v$ if and only if $c_\pi + \rho s_\pi \geq v$ for all $\pi \in \mathcal{B}$, and also if and only if this holds for all balanced mixtures of at most $m + 1$ points. A mixture with $s_\pi = 0$ gives positive

weight only to points of F , so $c_\pi \geq v$ and its condition always holds. For $s_\pi > 0$ the condition reads $\rho \geq (v - c_\pi)/s_\pi$. Since $D_\rho \leq v$ always, $D_\rho = v$ holds exactly when $\rho \geq 0$ is at least every such ratio. If the supremum of the ratios is less than $+\infty$, the least such ρ is (4), and it satisfies all the conditions; if the supremum is $+\infty$, no ρ does. $\square$

Each condition $c_\pi + \rho s_\pi \geq v$ is closed and monotone in ρ . Hence, when ρ_* is finite, the value-exact coefficients form the interval $[\rho_*, \infty)$, just as the coefficients exact at λ form $[\rho_*(\lambda), \infty)$. In words, ρ_* is the supremum of the objective gain $v - c_\pi$ per unit of average residual norm over balanced mixtures with $s_\pi > 0$, or zero if this supremum is negative or no such mixture exists. In (2) the ratio is taken over single points instead, and it includes the multiplier term.

2.4 One relaxed equality

For one relaxed equality the least penalty and the thresholds have closed forms in terms of two one-sided rates.

Proposition 2.4 (One relaxed equality). *Let $m = 1$ with the absolute value as norm, and suppose that X contains points with $r(x) > 0$ and points with $r(x) < 0$. Define*

$$A_+ = \sup_{x \in X: r(x) > 0} \frac{v - f(x)}{r(x)}, \quad A_- = \sup_{x \in X: r(x) < 0} \frac{v - f(x)}{-r(x)}.$$

Then $\rho_(\lambda) = \max\{0, A_+ - \lambda, A_- + \lambda\}$ for every $\lambda \in \mathbb{R}$; in particular, the zero-multiplier threshold is $\rho_*(0) = \max\{0, A_+, A_-\}$. If A_+ and A_- are finite, then for all $\rho \geq 0$ and $\lambda \in \mathbb{R}$,*

$$\rho_* = \max\{0, (A_+ + A_-)/2\}, \quad L_\rho(\lambda) = v \iff A_+ - \rho \leq \lambda \leq \rho - A_-. \quad (5)$$

If A_+ or A_- is $+\infty$, no finite ρ is value-exact.

Proof. The fixed-multiplier part is (2) split by the sign of the residual; the optimized part follows once we show that an exact dual value is attained.

For $r(x) > 0$ the ratio in (2) is $(v - f(x))/r(x) - \lambda$, and for $r(x) < 0$ it is $(v - f(x))/(-r(x)) + \lambda$. Their suprema are $A_+ - \lambda$ and $A_- + \lambda$, which gives the formula for $\rho_*(\lambda)$. Equivalently, $L_\rho(\lambda) = v$ if and only if

$$(\rho + \lambda)r(x) \geq v - f(x) \text{ whenever } r(x) > 0, \quad (\rho - \lambda)|r(x)| \geq v - f(x) \text{ whenever } r(x) < 0.$$

When A_+ and A_- are finite, these conditions are the interval in (5).

Next, if $D_\rho = v$, some multiplier attains it. Fix points x_+ and x_- with $r(x_+) > 0$ and $r(x_-) < 0$. Then $L_\rho(\lambda) \leq f(x_+) + (\lambda + \rho)r(x_+)$, which tends to $-\infty$ as $\lambda \rightarrow -\infty$, and $L_\rho(\lambda) \leq f(x_-) + (\rho - \lambda)|r(x_-)|$, which tends to $-\infty$ as $\lambda \rightarrow +\infty$. Hence every sequence λ_j with $L_\rho(\lambda_j) \rightarrow v$ is bounded. A subsequence converges to some $\bar{\lambda}$, and continuity of L_ρ gives $L_\rho(\bar{\lambda}) = v$.

Therefore $D_\rho = v$ if and only if $\rho \geq \max\{0, A_+ - \lambda, A_- + \lambda\}$ for some λ . If A_+ or A_- is $+\infty$, this never holds. Otherwise the interval in (5) is nonempty exactly when $A_+ - \rho \leq \rho - A_-$, that is, when $\rho \geq (A_+ + A_-)/2$. Together with $\rho \geq 0$ this gives ρ_* . $\square$

In words: a point with positive residual is charged $\rho + \lambda$ per unit of residual, and a point with negative residual is charged $\rho - \lambda$. A multiplier moves charge from one side to the other but cannot change the total 2ρ . The two sides need at least A_+ and A_- per unit, respectively. So the optimized threshold is the average $(A_+ + A_-)/2$ of the two one-sided rates, while multiplier zero needs their maximum. A rate can be negative: $A_+ < 0$ means that, for some $c > 0$, every

point with $r(x) > 0$ costs at least $v + cr(x)$, and similarly for A_- . When both are nonnegative, $\rho_*(0) = \max\{A_+, A_-\} \leq A_+ + A_- = 2\rho_*$: optimizing the multiplier reduces the required coefficient by at most a factor two. When $(A_+ + A_-)/2 \geq 0$, the interval in (5) at $\rho = \rho_*$ consists of the single multiplier $\lambda = (A_+ - A_-)/2$.

For $m = 1$, if residuals occur on only one side of zero, every balanced mixture is supported on F : its weighted residuals have one sign and sum to zero. Hence $c_\pi \geq v$ for every $\pi \in \mathcal{B}$, so Proposition 2.2 gives $D_\rho = v$ for all $\rho \geq 0$, and $\rho_* = 0$. Attainment can nevertheless fail.

Example 2.5 (Value-exact, but exact at no multiplier). Let $X = [0, 1]$, $r(x) = x$ and $f(x) = -\sqrt{x}$, with the absolute value as norm. Then $F = \{0\}$ and $v = 0$. All residuals are nonnegative, so $D_\rho = 0 = v$ for every $\rho \geq 0$. Fix λ and ρ , and put $t = \lambda + \rho$. For every $x \in (0, 1]$ with $tx < \sqrt{x}$, which holds for all small $x > 0$, the penalized objective $-\sqrt{x} + tx$ is negative. Hence $L_\rho(\lambda) < 0 = v$ for every finite λ and ρ : the threshold $\rho_*(\lambda)$ is $+\infty$ for every λ , although $\rho_* = 0$. Appendix A records the same phenomenon in a mixed-integer example with a convex quadratic constraint from the literature.

2.5 A sufficient condition from integer slices

The next lemma is the template for the upper bounds of Sections 4 and 5. It reduces exactness at multiplier zero to two separate quantities: a multiplier bound on each equality-feasible slice and the residual separation of each equality-infeasible slice.

Lemma 2.6 (A bound from integer slices). *Suppose that X is the union of finitely many nonempty compact sets X_z (for example, the nonempty slices of a mixed-integer native set). Let $\underline{f}$ and $\bar{f}$ be numbers with $f \geq \underline{f}$ on X and $v \leq \bar{f}$, and let $\Lambda \geq 0$. Suppose that for every X_z that meets F there is $\lambda_z \in \mathbb{R}^m$ with $\|\lambda_z\|_* \leq \Lambda$ and*

$$f(x) + \lambda_z^T r(x) \geq v_z := \min_{x' \in X_z \cap F} f(x') \quad \text{for all } x \in X_z,$$

and that some $\delta > 0$ satisfies $\min_{x \in X_z} \|r(x)\| \geq \delta$ for every X_z that does not meet F . Then every ρ with

$$\rho \geq \max\{\Lambda, (\bar{f} - \underline{f})/\delta\} \tag{6}$$

is exact at multiplier zero, and every ρ strictly larger than this maximum is minimizer-exact at multiplier zero. If every X_z meets F , the second term in (6) may be omitted.

Proof. Let ρ satisfy (6) and let $x \in X_z$. If X_z meets F , then $\lambda_z^T r(x) \leq \|\lambda_z\|_* \|r(x)\| \leq \rho \|r(x)\|$, so

$$f(x) + \rho \|r(x)\| \geq f(x) + \lambda_z^T r(x) \geq v_z \geq v.$$

If X_z does not meet F , then $f(x) + \rho \|r(x)\| \geq \underline{f} + \rho\delta \geq \bar{f} \geq v$. Thus $L_\rho(0) \geq v$, and weak duality gives $L_\rho(0) = v$. Hence $\rho_*(0)$ is at most the maximum in (6), and every strictly larger coefficient is minimizer-exact at multiplier zero by the consequences of (2). $\square$

The two terms in (6) control different parts of the penalty. The multiplier bound Λ concerns equality-feasible slices: since $f(x) \geq v_z - \Lambda \|r(x)\|$ on such a slice, it bounds how fast the objective can improve when the slice leaves the relaxed equalities. The separation δ concerns equality-infeasible slices: it bounds how close a slice that cannot satisfy the relaxed equalities comes to satisfying them, which matters when such a slice contains improving points. When f and the native constraints are convex on each slice and r is affine, Slater-type regularity on equality-feasible slices, such as the refined Slater condition of Assumption 4.1, supplies the multipliers λ_z and hence a finite Λ ; it says

nothing about δ . The chain of Section 3 shows that δ can be the bottleneck: its slices have strict points with slack $1/8$ and $\lambda_z = 0$ works on its equality-feasible slice, yet its equality-infeasible slice has separation $\delta_d = 2^{-2^d}$, and the lemma's coefficient $1/\delta_d$ is exactly its zero-multiplier threshold.

3 A doubly exponential lower bound

This section shows that convex quadratic native constraints can force a doubly exponentially large exact penalty. We construct instances with one binary variable, $d + 1$ continuous variables and $O(d \log(d + 1))$ input bits whose least penalty is $2^{2^d - 1}$, so that every exact rational coefficient needs at least 2^d numerator bits. For a convex quadratic objective over a rational mixed-integer polyhedron, some exact norm penalty has polynomial encoding length [22, Theorem 11]; the construction, whose objective is linear, shows that this guarantee does not extend to convex quadratic native constraints. Numbers are encoded as described in Section 2.

3.1 Construction and intuition

For an integer $d \geq 1$, let

$$A_d = \{a \in [0, 1]^d : a_1 \geq 1/4, a_{i+1} \geq a_i^2 \ (1 \leq i < d)\}, \quad \delta_d = 2^{-2^d}. \quad (7)$$

Squaring is increasing on $[0, 1]$, so induction gives $a_i \geq 2^{-2^i}$ for every $a \in A_d$ and every i . The point $\bar{a}$ with $\bar{a}_i = 2^{-2^i}$ lies in A_d and attains all these bounds. In particular, the smallest value of a_d on A_d is δ_d . This squaring chain is classical (Section 1.2). Define

$$X_d = \{(a, y, q) : a \in A_d, -1 \leq y \leq 1, q \in \{0, 1\}, y \geq a_d - 2(1 - q)\}. \quad (8)$$

We call the last inequality the *switching inequality*. The instance minimizes $-q$ over $(a, y, q) \in X_d$ subject to the relaxed equality $y = 0$. In the notation of (1), $X = X_d$, $f = -q$, $r = y$, $m = 1$ and $\|r\| = |y|$. We write v_d , $L_{d,\rho}(\lambda)$, $D_{d,\rho}$, $\rho_{*,d}$ and $\rho_{*,d}(\lambda)$ for the optimal value, the penalized value, the dual value, the least penalty and the threshold at multiplier λ of this instance.

Why the penalty must be large. Setting $q = 1$ lowers the objective by one. At $q = 1$ the switching inequality reads $y \geq a_d$, and the chain forces $a_d \geq \delta_d$. So the improving slice $q = 1$ never meets the relaxed equality $y = 0$, but it comes within distance δ_d of it, and this distance is attained. At multiplier zero, the penalty must charge the residual δ_d at least the objective gain of one, which requires $\rho \geq 1/\delta_d$. A positive multiplier λ helps at the improving point, because it adds $\lambda\delta_d$ to its penalized value. At $q = 0$, however, the switching inequality is slack, and y ranges over all of $[-1, 1]$ at cost zero. The point $(q, y) = (0, -1)$ has penalized value $\rho - \lambda$, so a multiplier larger than ρ creates a new violation there. Balancing the two points gives the threshold $1/(2\delta_d)$, half the zero-multiplier value. Residuals of both signs are therefore what make the bound hold for every multiplier. In the terms of Lemma 2.6, δ_d is the separation of the equality-infeasible slice $q = 1$, and the large coefficient is caused by this separation alone.

Since f and r depend only on (q, y) , all quantities above are determined by the projection of X_d onto the (q, y) -plane. The proof below computes this projection first.

3.2 The least penalty

Theorem 3.1. *Let $d \geq 1$. The instance defined by (8), with objective $-q$ and relaxed equality $y = 0$, has the following properties.*

(a) Its optimal value is $v_d = 0$, and for every $\rho \geq 0$

$$D_{d,\rho} = \min \left\{ 0, \frac{2\rho\delta_d - 1}{1 + \delta_d} \right\}. \quad (9)$$

For every $\rho \geq 0$ the dual supremum is attained by some multiplier.

(b) Its least penalty is

$$\rho_{*,d} = \frac{1}{2\delta_d} = 2^{2^d-1}. \quad (10)$$

Each notion of exactness in Definition 2.1 implies value-exactness, so every coefficient that is exact in any of these senses is at least $\rho_{*,d}$. If such a coefficient is a rational number ρ_1/ρ_2 with positive integers ρ_1, ρ_2 , then the numerator ρ_1 has at least 2^d bits.

(c) The instance has one binary variable, $d+1$ continuous variables and a linear objective. Its native constraints are the $d-1$ convex quadratic inequalities $a_i^2 \leq a_{i+1}$ and affine constraints, with all coefficients from a fixed finite set of rationals. Its sparse encoding length is $O(d \log(d+1))$.

(d) Each of the two integer slices $q = 0$ and $q = 1$ contains a point at which every native inequality in the continuous variables, the variable bounds included, holds with slack at least $1/8$. On the equality-feasible slice $q = 0$, this point also satisfies $y = 0$.

For $d = 1$ the chain has no quadratic inequality, so the instance is an MILP with $\rho_{*,1} = 2$; the convex quadratic structure starts at $d = 2$.

Proof. We compute the projection of X_d onto (q, y) , bound $D_{d,\rho}$ from above by a balanced mixture and from below by explicit multipliers, and then check parts (b)–(d) directly.

Projection. We show that

$$\{(q, y) : (a, y, q) \in X_d\} = (\{0\} \times [-1, 1]) \cup (\{1\} \times [\delta_d, 1]). \quad (11)$$

(i) If $q = 1$, the switching inequality and the chain give $y \geq a_d \geq \delta_d$, so the part with $q = 1$ lies in $\{1\} \times [\delta_d, 1]$. (ii) Conversely, take $a = \bar{a}$. At $q = 1$ the switching inequality reads $y \geq \delta_d$, so every $y \in [\delta_d, 1]$ occurs. At $q = 0$ it reads $y \geq a_d - 2$, and $a_d - 2 \leq -1$, so every $y \in [-1, 1]$ occurs. (iii) Since $\delta_d > 0$, a native point with $y = 0$ has $q = 0$, and such points exist. Hence $v_d = 0$.

Reduction to the projection. Since $f = -q$ and $r = y$, the penalized value is the minimum of $-q + \lambda y + \rho|y|$ over the set (11). For fixed q this function is piecewise linear in y with a single break at $y = 0$, so on each interval its minimum is attained at an endpoint or at $y = 0$.

Upper bound. The points $(q, y) = (0, -1)$ and $(1, \delta_d)$ have residuals -1 and δ_d . With weights $\delta_d/(1 + \delta_d)$ and $1/(1 + \delta_d)$ their average residual is zero, so they form a balanced mixture (Proposition 2.2). Their penalized values are $\rho - \lambda$ and $-1 + (\rho + \lambda)\delta_d$, and the weighted average is

$$\frac{\delta_d(\rho - \lambda) - 1 + (\rho + \lambda)\delta_d}{1 + \delta_d} = \frac{2\rho\delta_d - 1}{1 + \delta_d},$$

independently of λ . The feasible point $(0, 0)$ has penalized value zero. Hence $L_{d,\rho}(\lambda)$ is at most the right-hand side of (9) for every λ , and so is $D_{d,\rho}$.

Lower bound and attainment. First let $0 \leq \rho \leq 1/(2\delta_d)$ and take

$$\lambda_\rho = \frac{1 + \rho(1 - \delta_d)}{1 + \delta_d}.$$

Then $\lambda_\rho \geq \rho$, since this inequality is equivalent to $2\rho\delta_d \leq 1$. At $q = 0$, the function $\lambda_\rho y + \rho|y|$ is nondecreasing on $[-1, 1]$, so its minimum is at $y = -1$, with value $\rho - \lambda_\rho = (2\rho\delta_d - 1)/(1 + \delta_d)$. At $q = 1$, the function $-1 + (\lambda_\rho + \rho)y$ is nondecreasing, so its minimum is at $y = \delta_d$, with value $-1 + (\lambda_\rho + \rho)\delta_d = (2\rho\delta_d - 1)/(1 + \delta_d)$. Both values equal the right-hand side of (9), which is nonpositive here. Next let $\rho \geq 1/(2\delta_d)$ and take $\lambda = \rho$. At $q = 0$ the penalized value is $2\rho y \geq 0$ for $y \geq 0$ and 0 for $y \leq 0$. At $q = 1$ it is $-1 + 2\rho y \geq -1 + 2\rho\delta_d \geq 0$. So $L_{d,\rho}(\rho) = 0$, again the right-hand side of (9). In both cases the upper bound is attained, which proves (a).

Part (b). By (9), $D_{d,\rho} = 0 = v_d$ exactly when $2\rho\delta_d \geq 1$, which gives (10). The integer 2^{2^d-1} has exactly 2^d bits. If $\rho_1/\rho_2 \geq 2^{2^d-1}$ with positive integers ρ_1, ρ_2 , then $\rho_1 \geq \rho_1/\rho_2 \geq 2^{2^d-1}$, so ρ_1 has at least 2^d bits.

Part (c). Each inequality $a_i^2 - a_{i+1} \leq 0$ is convex quadratic. All other native constraints are affine: the bounds, the seed $a_1 \geq 1/4$ and the switching inequality $y - a_d - 2q \geq -2$. All coefficients and right-hand sides lie in $\{0, \pm 1, \pm 2, 1/4\}$. There are $O(d)$ nonzero coefficients, each stored with variable indices of $O(\log(d+1))$ bits.

Part (d). Set $a_i = 3/8$ for all i . The seed has slack $1/8$, each chain inequality has slack $3/8 - 9/64 = 15/64$, and the bounds $0 \leq a_i \leq 1$ have slack at least $3/8$. On the slice $q = 0$ take $y = 0$: the bounds on y have slack 1, the switching inequality has slack $0 - (3/8 - 2) = 13/8$, and the relaxed equality holds. On the slice $q = 1$ take $y = 1/2$: the bounds on y have slack $1/2$ and the switching inequality has slack $1/2 - 3/8 = 1/8$. The smallest slack is $1/8$ for every d , even though the slice $q = 1$ comes within δ_d of $y = 0$. $\square$

Multipliers and the zero-multiplier threshold. The threshold (2) at a fixed multiplier λ is

$$\rho_{*,d}(\lambda) = \max\{0, 1/\delta_d - \lambda, \lambda\}.$$

Indeed, the native points with $q = 1$ have ratios $(1 - \lambda y)/y = 1/y - \lambda$, largest at $y = \delta_d$; those with $q = 0$ and $y > 0$ have ratio $-\lambda$, and those with $y < 0$ have ratio λ . Equivalently, Proposition 2.4 applies with $A_+ = 1/\delta_d$ and $A_- = 0$. This formula gives four facts.

- $\lambda = \rho_{*,d}$ is the unique multiplier at which $\rho_{*,d}$ is exact.
- At $\rho = \lambda = \rho_{*,d}$, both points $(q, y) = (0, -1)$ and $(1, \delta_d)$ have penalized value $0 = v_d$. So $\rho_{*,d}$ is exact at this multiplier but not minimizer-exact.
- For every $\rho > \rho_{*,d}$, the pair $(\rho, \rho_{*,d})$ is minimizer-exact: every native point with $y \neq 0$ has positive penalized value, namely $(\rho + \rho_{*,d})y$ or $(\rho - \rho_{*,d})|y|$ at $q = 0$, and at least $-1 + (\rho + \rho_{*,d})\delta_d > 0$ at $q = 1$.
- The zero-multiplier threshold is $\rho_{*,d}(0) = 1/\delta_d = 2^{2^d}$, twice the least penalty; it has $2^d + 1$ bits.

Thus optimizing the multiplier saves exactly one bit here. Appendix B gives a variant with two binary variables in which residuals of both signs occur at cost -1 ; there the optimal multiplier is zero and both thresholds equal 2^{2^d} .

Remark 3.2 (Assumptions of Lefebvre and Schmidt). The instance satisfies Assumptions 1–5 of Lefebvre and Schmidt [29]. Write $q \in \{0, 1\}$ as an integer variable with bounds $0 \leq q \leq 1$ and take the absolute value as penalty function. Assumptions 1–4 are then immediate: X_d is nonempty and compact, the objective $-q$ is linear, and the penalty function is a norm. Their Assumption 5, a Slater condition for each fixed integer assignment, holds at $q = 0$ through the strict point with $a_i = 3/8$ and $y = 0$ of Theorem 3.1(d), and at $q = 1$ because no native point satisfies $y = 0$ (since

$y \geq \delta_d > 0$). Their Theorem 14 therefore gives, for every fixed multiplier, a finite penalty that is exact at that multiplier; here this is the finite threshold $\rho_{*,d}(\lambda)$ above. Theorem 3.1 shows that these penalties need at least 2^d bits, even after the multiplier is optimized.

3.3 Encoding consequences

The lower bound survives if only a fixed additive accuracy is required.

Corollary 3.3 (Fixed additive accuracy). *Let $0 \leq \epsilon < 1$. For the instance of Theorem 3.1, $v_d - D_{d,\rho} \leq \epsilon$ holds if and only if*

$$\rho \geq \max \left\{ 0, \frac{1 - \epsilon(1 + \delta_d)}{2\delta_d} \right\}. \quad (12)$$

For each fixed $\epsilon < 1$ and all sufficiently large d , the numerator of every rational ρ satisfying (12), written with positive integer numerator and denominator, has at least $2^d - O_\epsilon(1)$ bits.

Proof. By (9) and $v_d = 0$, the gap is $\max\{0, (1 - 2\rho\delta_d)/(1 + \delta_d)\}$. Since $\epsilon \geq 0$, it is at most ϵ exactly when $1 - 2\rho\delta_d \leq \epsilon(1 + \delta_d)$, which is (12). If d is so large that $\epsilon\delta_d \leq (1 - \epsilon)/2$, then $1 - \epsilon - \epsilon\delta_d \geq (1 - \epsilon)/2$, and the right-hand side of (12) is at least $(1 - \epsilon)/(4\delta_d) = (1 - \epsilon)2^{2^d - 2} > 0$. As in Theorem 3.1(b), the numerator is then at least $(1 - \epsilon)2^{2^d - 2}$, so it has more than $2^d - 2 + \log_2(1 - \epsilon)$ bits. $\square$

For example, accuracy $\epsilon = 1/2$ requires $\rho \geq (1 - \delta_d)/(4\delta_d)$. The objective takes only the values 0 and -1 , so its range is one, and ϵ is a fixed accuracy relative to that range. For $\epsilon \geq 1$ the condition holds for every $\rho \geq 0$, because the gap never exceeds $1/(1 + \delta_d)$.

Remark 3.4 (Input length). The sparse encoding length N of the instance is $\Theta(d \log(d + 1))$, because it records $\Theta(d)$ variable indices, most of which need $\Theta(\log(d + 1))$ bits. Hence $d = \Theta(N/\log N)$, and the lower bound of 2^d bits is $2^{\Theta(N/\log N)}$ bits. This is superpolynomial in N but not $2^{\Omega(N)}$. In the continuous dimension $n = d + 1$ the bound is exponential. Remark 3.5 below adds a linear dependence on N . The bound concerns the size of exact penalties only. It is not a lower bound on the effort to solve the original problem, whose value $v_d = 0$ is immediate from (11).

Remark 3.5 (A longer seed). Fix an integer $t \geq 2$ and replace the seed inequality $a_1 \geq 1/4$ in (7) by $a_1 \geq 2^{-t}$; the case $t = 2$ is the instance of Theorem 3.1. The same induction gives $a_i \geq 2^{-t2^{i-1}}$, with equality at $\bar{a}_i = 2^{-t2^{i-1}}$, so the separation of the slice $q = 1$ becomes $\delta = 2^{-t2^{d-1}}$. The proof of Theorem 3.1 uses only that the chain forces $a_d \geq \delta$, that $\bar{a}$ attains $a_d = \delta$, and that $0 < \delta \leq 1$. It therefore applies with δ in place of δ_d . The least penalty is $1/(2\delta) = 2^{t2^{d-1}-1}$, which has $t2^{d-1}$ bits, and the zero-multiplier threshold $1/\delta$ has $t2^{d-1} + 1$ bits. The strict points with $a_i = 3/8$ remain valid: the seed slack is $3/8 - 2^{-t} \geq 1/8$, and the other slacks do not change. The sparse encoding length becomes $N = O(d \log(d + 1) + t)$, and $N > t$ because the seed coefficient alone has $t + 1$ bits.

Now take t at least of order $d \log(d + 1)$, so that $N = \Theta(t)$. The instance has $n = d + 1$ continuous variables, satisfies Assumption 4.1, and every exact rational coefficient has a numerator of at least $t2^{d-1} = \Omega(N2^n)$ bits. Since the chain has $k = d - 1$ native inequalities that are nonlinear in the continuous variables, this is also $\Omega(N2^k)$ bits. Sections 4 and 5 compare these lower bounds with the upper bounds.

3.4 What the bound depends on

Theorem 3.1 concerns a fixed norm penalty, a fixed relaxed equality and ordinary binary encoding of the coefficient. The penalty function matters only through its linear growth at zero; changing the growth, the scaling of the equality or the number format changes the question.

Penalty functions with at most linear growth. Let $\psi : \mathbb{R} \rightarrow [0, \infty)$ satisfy $\psi(y) \leq C|y|$ for $|y| \leq 1$, with a constant $C > 0$, and replace $\rho|y|$ by $\rho\psi(y)$ in the penalized value. The balanced mixture from the proof of Theorem 3.1, with the same weights, gives for every multiplier λ the upper bound

$$\frac{-1 + \rho(\psi(\delta_d) + \delta_d\psi(-1))}{1 + \delta_d} \leq \frac{-1 + 2C\rho\delta_d}{1 + \delta_d}.$$

So the dual value is negative, and the penalty is not value-exact, whenever $\rho < 1/(2C\delta_d)$. This includes the squared penalty $\psi(y) = y^2$ of the classical augmented Lagrangian, with $C = 1$. The obstruction therefore comes from linear growth at zero, not from the norm axioms.

Fractional powers. Without linear growth the threshold can be small. Let $p_d = 2^{-d}$, a rational exponent with $O(d)$ bits. Then $\delta_d^{p_d} = 1/2$. With multiplier zero and augmentation $2|y|^{p_d}$, the penalized objective $-q + 2|y|^{p_d}$ is nonnegative at $q = 0$, with value zero at $y = 0$, and at $q = 1$ it is at least $-1 + 2\delta_d^{p_d} = 0$. So the coefficient two is exact, and every larger coefficient is minimizer-exact at multiplier zero. This augmenting function, however, depends on the instance, is not Lipschitz at zero, is not a norm and is not convex. It changes both the encoding question and the continuous subproblem.

Scaling and number format. Rescaling the relaxed equality to $2^{2^d}y = 0$ divides the least penalty and the zero-multiplier threshold by 2^{2^d} , so the least penalty becomes $1/2$. The doubly exponential number then sits in the data: the coefficient 2^{2^d} has $2^d + 1$ bits, so the input length becomes exponential in d . Similarly, writing coefficients in a succinct exponent notation changes the representation whose length Theorem 3.1 measures. Both changes alter the formulation or representation to which the theorem applies.

4 Encoding upper bounds in the continuous dimension

This section proves that the doubly exponential growth of Section 3 is the worst possible in the continuous dimension, up to the constant in the exponent. For bounded mixed-integer quadratic models that are convex in the continuous variables for each integer assignment and satisfy the refined Slater condition (Assumption 4.1), some integer coefficient with $(N + 1)2^{O(n)}$ bits is minimizer-exact at multiplier zero, where N is the input length and n the number of continuous variables. The bound does not depend on the number of integer variables, integer assignments or affine rows, except through N .

4.1 Model and assumptions

Throughout this section the instance is

$$v = \min\{f(x, z) : (x, z) \in X, Ax + Bz - b = 0\}, \quad x \in \mathbb{R}^n, z \in \mathbb{Z}^p, \quad (13)$$

with native set $X = \{(x, z) \in \mathbb{R}^n \times \mathbb{Z}^p : g_i(x, z) \leq 0 \ (1 \leq i \leq s)\}$ and relaxed equalities $r(x, z) = Ax + Bz - b = 0$, where A, B and b are rational, $n \geq 1$ and r has $m \geq 1$ coordinates. Since r links the continuous and the integer variables, we also call these equalities the linking equalities and their multipliers linking multipliers. The number s counts all native inequalities, including the variable bounds; native affine equalities are written as pairs of inequalities. We call each native inequality a *row*. For an integer assignment z we identify the slice with $X_z = \{x \in \mathbb{R}^n : (x, z) \in X\}$; equality-feasible and equality-infeasible slices are as in Section 2, and empty slices are discarded.

Assumption 4.1. The instance (13) satisfies the following.

- (A1) *Quadratic data in explicit encoding.* The objective f and the native constraint functions $g_1, \dots, g_s$ are polynomials in (x, z) of total degree at most two with rational coefficients. The input lists them in expanded form, with every coefficient and every variable index written in binary, together with A , B and b . Its total binary length is $N \geq 2$; in particular $s, m, n \leq N$.
- (A2) *Finite bounds.* Every continuous and every integer variable has finite rational lower and upper bounds, listed among the native inequalities.
- (A3) *Convexity in the continuous variables.* For every integer assignment z , the functions $f(\cdot, z)$ and $g_i(\cdot, z)$, $1 \leq i \leq s$, are convex on $\mathbb{R}^n$.
- (A4) *Feasibility and refined Slater condition.* The feasible set of (13) is nonempty, and every equality-feasible slice X_z contains a point x with $Ax + Bz = b$ and $g_i(x, z) < 0$ for every native inequality g_i that is nonlinear in x .

By (A2), X is compact and there are finitely many slices. The x -Hessian of a polynomial of degree at most two is constant. Hence convexity in (A3), and whether a row is nonlinear in x , do not depend on z . In (A4), inequalities affine in x may hold with equality; this includes rows such as $x_1 z_1 + z_1^2 \leq 3$ that contain integer variables. The condition asks for no quantitative margin, and nothing is assumed on equality-infeasible slices. This matters: in the chain of Section 3, the large penalty comes from an equality-infeasible slice.

For a positive integer a , write $\text{bit}(a) = 1 + \lfloor \log_2 a \rfloor$ for its binary length. All constants implicit in $O(\cdot)$ below are absolute. In particular, they do not depend on the number p of integer variables or on the number of slices; the bounds depend on the instance only through N and n .

4.2 Main result and proof strategy

Theorem 4.2. Under Assumption 4.1, there is an integer $\rho \geq 1$ with

$$\text{bit}(\rho) \leq (N + 1)2^{O(n)} \quad (14)$$

that is minimizer-exact at multiplier zero for both the infinity norm and the one-norm; the same ρ works for both.

Minimizer-exactness at multiplier zero implies exactness at multiplier zero and value-exactness (Section 2). Hence both thresholds $\rho_*(0)$ and ρ_* are at most this ρ .

Proof strategy. We use the infinity norm, whose dual norm is the one-norm, and deduce the one-norm case at the end. Lemma 2.6 reduces exactness at multiplier zero to two uniform bounds:

- (a) on each equality-feasible slice, one linking multiplier λ_z with $f + \lambda_z^T r \geq v_z$ on the slice and a bound on its dual norm $\|\lambda_z\|_1$;
- (b) on each equality-infeasible slice, a lower bound on $\min_{x \in X_z} \|r(x, z)\|_\infty$.

After z is substituted, both become bounds on the coordinates of points of basic closed semialgebraic sets in $O(n)$ variables, defined by polynomials of degree two with integer coefficients of bitsize $O(N)$. One result from real algebraic geometry (Lemma 4.3) bounds such coordinates by a radius R with $\log R = O(N)2^{O(n)}$. The radius depends only on the number of variables, the number of

polynomials and the coefficient bitsize. It is therefore the same for every slice, and no factor for the number of slices appears.

The number of variables is the critical parameter, because the radius bound is exponential in it. For (a), the full KKT system has $n + s + m$ unknowns, which would give $2^{O(n+s+m)}$ in the exponent. A conic Carathéodory argument keeps at most n multipliers, so the system has at most $2n$ variables. For (b), Lemma 4.3 bounds coordinates of points, not distances from zero; we therefore encode the reciprocal $1/\|r\|_\infty$ as one extra variable.

4.3 Effective radii

A *basic closed set* is the set of real solutions of a finite conjunction of polynomial equations $P = 0$ and weak inequalities $P \leq 0$ or $P \geq 0$.

Lemma 4.3 (Effective radii). *Let $\mathcal{P}$ be a list of at most S polynomials in ℓ variables, with integer coefficients of bitsize at most $\tau \geq 1$ and degree at most two. There is a radius $R \geq 1$, depending only on (S, ℓ, τ) , with*

$$\log_2 R \leq (\tau + \log_2(S + 1) + 1)2^{O(\ell)}, \quad (15)$$

such that the closed ball of radius R centred at the origin meets every nonempty basic closed set defined by polynomials from $\mathcal{P}$ and contains every such set that is bounded. The same holds for finite unions of such sets.

Proof. We combine two radius bounds of Basu and Roy [5], stated for weak sign conditions. A weak sign condition on $\mathcal{P}$ prescribes, for each $P \in \mathcal{P}$, one of $P = 0$, $P \leq 0$ or $P \geq 0$; its realization is the set of points of $\mathbb{R}^\ell$ that satisfy all these conditions. Apply the bounded-component radius of Basu and Roy [5, Theorem 3] and the component-meeting radius of Basu and Roy [5, Theorem 4] to weak sign conditions on the full list of polynomials and take the larger radius, or 1 if that is larger. The first ball contains every bounded connected component of every realization; the second meets every connected component of every nonempty realization.

These theorems use auxiliary parameters called N and D ; to avoid a clash with the input length, we write them N_{BR} and D_{BR} . For degree two, $N_{\text{BR}} = 5 \cdot 4^{\ell-1}$ and $D_{\text{BR}} = 3\ell + 2$ in Theorem 3, and $N_{\text{BR}} = 6 \cdot 5^{\ell-1}$ and $D_{\text{BR}} = 4\ell + 2$ in Theorem 4. Theorem 4 rests on Basu and Roy [5, Theorem 2], whose proof contains a term $\text{bit}(N_{\text{BR}}!)$ in its bound on $1/u_0$ that the statements omit; we retain it. Substituting these values into the explicit formulas shows that the binary logarithm of each radius is at most $2^{O(\ell)}(\tau + \text{bit}(S) + 1)$; since $N_{\text{BR}} = 2^{O(\ell)}$, the term $\text{bit}(N_{\text{BR}}!)$ is also absorbed by $2^{O(\ell)}$. This gives (15), because $\text{bit}(S) \leq \log_2(S + 1) + 1$. The formulas depend only on the number of polynomials, the number of variables, the degree and the coefficient bitsize, and they do not decrease when the number of polynomials grows. Hence R depends only on (S, ℓ, τ) .

Let Y be a basic closed set defined by polynomials from $\mathcal{P}$. It is the union of the realizations of those weak sign conditions on the full list that are compatible with its definition; a polynomial not used in the definition contributes the two choices $P \leq 0$ and $P \geq 0$. If Y is nonempty, one of these realizations is nonempty, and the ball meets each of its connected components; hence it meets Y . If Y is bounded, each of these realizations is bounded, so all of its connected components are bounded and lie in the ball; hence Y lies in the ball. For a finite union, apply the first argument to a nonempty member; if the union is bounded, every member is bounded, and the second argument applies to each member. $\square$

4.4 Uniform coefficient heights

Lemma 4.3 needs integer coefficients of bounded bitsize. All polynomial systems below are built from the input polynomials after z is substituted, by differentiation in x and by multiplying affine polynomials by one new variable (a multiplier, or the variable t of Section 4.6). We check that each such system can be written with integer coefficients of bitsize

$$\tau = O(N), \quad (16)$$

uniformly over all integer assignments.

- Multiply every equation and inequality of a system by the product of all denominators in the input. This positive integer is below 2^N and turns every input coefficient into an integer of absolute value below 2^N . Multiplying an equation or inequality by a positive integer does not change its solution set, so all variables, including the multipliers, keep their meaning. We do not rescale the residual r in the penalty itself, since that would change the penalty.
- By (A2), the bounds on all variables are input inequalities, so every coordinate of a point of X , continuous or integer, has absolute value below 2^N ; in particular $|z_j| < 2^N$. Substituting z into a monomial of degree at most two multiplies its coefficient by less than 2^{2N} . Each polynomial has at most N monomials, so collecting terms adds at most $\log_2 N + 1$ bits.
- Differentiation multiplies coefficients by at most two, and multiplying a polynomial by a new variable leaves its coefficients unchanged.

We also use an objective bound $|f| \leq F_0$ on X , with an integer $F_0 \geq 1$ and $\text{bit}(F_0) = O(N)$. For example, let c_f be the sum of the absolute values of the objective coefficients and take $F_0 = 1 + \lceil c_f 2^{2N} \rceil$. Then $|f| \leq F_0$ on X , because every coordinate is below 2^N in absolute value; since $c_f < N 2^N$, we get $\text{bit}(F_0) = O(N)$.

4.5 Feasible slices: a sparse KKT certificate

Fix an equality-feasible assignment z and suppress it from the notation: $g_i(x) = g_i(x, z)$, $r(x) = Ax + Bz - b$, and

$$v_z = \min\{f(x) : x \in X_z, r(x) = 0\}.$$

We need a linking multiplier $\lambda_z \in \mathbb{R}^m$ with

$$f(x) + \lambda_z^T r(x) \geq v_z \quad \text{for all } x \in X_z, \quad (17)$$

and a bound on $\|\lambda_z\|_1$. A KKT multiplier satisfies (17). The work is to find one that is a coordinate of a point in a system with at most $2n$ variables.

Existence of a KKT certificate. The slice problem has an optimal solution x^* , because its feasible set is nonempty and compact. By (A3) and (A4), it is a convex program with finite convex functions on $\mathbb{R}^n$ and affine equality constraints that has a feasible point satisfying strictly every inequality that is not affine. Hence Rockafellar [31, Theorem 28.2] gives a Kuhn–Tucker vector, and for differentiable data Rockafellar [31, Theorem 28.3] gives the KKT conditions at x^* : there are $\mu \in \mathbb{R}^s$ and $\lambda \in \mathbb{R}^m$ with

$$\mu \geq 0, \quad \mu_i g_i(x^*) = 0 \quad (1 \leq i \leq s), \quad \nabla f(x^*) + \sum_{i=1}^s \mu_i \nabla g_i(x^*) + A^T \lambda = 0.$$

This holds even when the active affine rows are linearly dependent.

Sparsification. Let $e = \text{rank } A$, and choose a set J of e linearly independent rows of A that span its row space W . Stationarity says that $-\nabla f(x^*)$ is a nonnegative combination of active native gradients plus a vector in W . Modulo W we are in the space $\mathbb{R}^n/W$ of dimension $n - e$. There, by the conic Carathéodory theorem, the class of $-\nabla f(x^*)$ is a nonnegative combination of the classes of at most $n - e$ active native gradients, indexed by a set I . The remainder lies in W and is therefore a combination of the rows in J . This gives new multipliers $\mu_i \geq 0$ ($i \in I$) and λ_j ($j \in J$) with $|I| + |J| \leq n$. The sets I and J depend on z , and I may contain variable-bound rows. Consequently the following system in (x, μ_I, λ_J) has a solution:

$$\begin{aligned} g_i(x) &\leq 0 \quad (1 \leq i \leq s), & Ax + Bz - b &= 0, \\ \mu_i &\geq 0, \quad g_i(x) &= 0 \quad (i \in I), \\ \nabla f(x) + \sum_{i \in I} \mu_i \nabla g_i(x) + \sum_{j \in J} \lambda_j A_j^T &= 0. \end{aligned} \tag{18}$$

All m linking rows remain as constraints, although only the multipliers indexed by J appear. The equations $g_i(x) = 0$ for $i \in I$ replace the complementarity products $\mu_i g_i(x) = 0$, which have degree three, and reuse the native polynomials. Hence (18) is a basic closed set in at most $2n$ variables, defined by at most $s + m + 2n$ polynomials of degree at most two (each $\mu_i \nabla g_i$ is a multiplier times an affine function).

Every solution is a certificate. Let (x, μ_I, λ_J) solve (18), and extend λ_J by zeros to $\lambda_z \in \mathbb{R}^m$. The Lagrangian

$$\mathcal{L}(y) = f(y) + \sum_{i \in I} \mu_i g_i(y) + \lambda_z^T r(y)$$

is convex on $\mathbb{R}^n$ and stationary at x , so it is minimized at x , where its value is $f(x)$. For every feasible y of the slice problem, $f(y) \geq \mathcal{L}(y) \geq \mathcal{L}(x) = f(x)$; since x is itself feasible, $f(x) = v_z$. For every $y \in X_z$, the terms $\mu_i g_i(y)$ are nonpositive, so $f(y) + \lambda_z^T r(y) \geq \mathcal{L}(y) \geq v_z$. This is (17).

Size of the certificate. By (16), the system (18) can be cleared to integer coefficients of bitsize $\tau = O(N)$. Let R_M be the largest of the radii that Lemma 4.3 gives for at most $s + m + 2n$ polynomials, at most $2n$ variables and bitsize τ . By the meeting part of the lemma, some solution of (18) has all coordinates bounded by R_M in absolute value, where

$$\log_2 R_M \leq (\tau + \log_2(s + m + 2n + 1) + 1)2^{O(n)}. \tag{19}$$

The corresponding multiplier satisfies $\|\lambda_z\|_1 \leq |J|R_M \leq nR_M$. The bound R_M is the same for every equality-feasible slice. It bounds one suitable certificate, not all optimal multipliers, whose set can be unbounded.

4.6 Infeasible slices: separation from the relaxed equalities

Fix a nonempty equality-infeasible slice X_z . Since X_z is compact and r does not vanish on it,

$$\delta_z = \min_{x \in X_z} \|Ax + Bz - b\|_\infty > 0.$$

We need a lower bound on δ_z , but Lemma 4.3 bounds coordinates of points. The idea is to encode $t = 1/\|r(x, z)\|_\infty$ as a coordinate: its graph over X_z is bounded, so the lemma bounds t , and hence $1/\delta_z$. The infinity norm keeps this description at degree two. For the Euclidean norm one would need $t^2\|r\|_2^2 = 1$, of degree four.

Write $r_j = r_j(x, z)$ and consider

$$\Gamma_z = \left\{ (x, t) : \begin{array}{l} x \in X_z, \quad t \geq 0, \quad -1 \leq tr_j \leq 1 \quad (1 \leq j \leq m), \\ tr_j = 1 \text{ or } tr_j = -1 \text{ for at least one } j \end{array} \right\}. \quad (20)$$

For $x \in X_z$, these conditions say $t|r_j| \leq 1$ for all j , with equality for some j ; thus $t = 1/\|r(x, z)\|_\infty$. Hence Γ_z is exactly the graph of $t = 1/\|r(x, z)\|_\infty$ over X_z . It is nonempty and compact, since $\|r\|_\infty \geq \delta_z > 0$ on the compact set X_z . Choosing a coordinate j and a sign in the last line writes Γ_z as a union of $2m$ basic closed sets. Each equation $tr_j = \pm 1$ uses a polynomial already present in $-1 \leq tr_j \leq 1$. So only $s + 2m + 1$ polynomials of degree two in $n + 1$ variables are needed, and by (16) their coefficients can be cleared to bitsize $\tau = O(N)$. The containment part of Lemma 4.3 bounds t on Γ_z . At a minimizer of $\|r\|_\infty$ over X_z , $t = 1/\delta_z$; therefore

$$\delta_z^{-1} \leq R_D, \quad \log_2 R_D \leq (\tau + \log_2(s + 2m + 2) + 1)2^{O(n)}, \quad (21)$$

where R_D is the radius of Lemma 4.3 for $s + 2m + 1$ polynomials, $n + 1$ variables and bitsize τ ; it is the same for every equality-infeasible slice. This step uses no KKT condition, which is why no regularity is needed on equality-infeasible slices.

4.7 Proof of the theorem and comparison with the lower bound

Proof of Theorem 4.2. We apply Lemma 2.6 with the bounds of the two preceding subsections. Let $E \geq 0$ be the least integer with $2^E \geq \max\{R_M, R_D\}$, where R_M and R_D are the uniform radii in (19) and (21), and set

$$\rho = (n + 2F_0 + 1)2^E. \quad (22)$$

This integer is strictly larger than both nR_M and $2F_0R_D$.

Consider first the infinity norm, whose dual norm is the one-norm. Apply Lemma 2.6 with objective bounds $\underline{f} = -F_0$ and $\bar{f} = F_0$, multiplier bound $\Lambda = nR_M$ from the feasible slices, and separation $\delta = 1/R_D$ from (21). Since ρ is strictly larger than $\max\{\Lambda, (\bar{f} - \underline{f})/\delta\} = \max\{nR_M, 2F_0R_D\}$, it is minimizer-exact at multiplier zero. In detail, on an equality-feasible slice a point with nonzero residual has penalized value at least $v_z + (\rho - nR_M)\|r\|_\infty > v$, by (17) and $|\lambda_z^T r| \leq \|\lambda_z\|_1 \|r\|_\infty$. On an equality-infeasible slice every point has penalized value at least $-F_0 + \rho/R_D > F_0 \geq v$.

The same coefficient works for the one-norm. Indeed, $\|r\|_1 \geq \|r\|_\infty$, and both vanish on the feasible set of (13). So the one-norm penalized objective is at least the infinity-norm one and equals f on the feasible set; its minimum value is v , and its minimizers are exactly the minimizers of (13).

Finally, $\text{bit}(\rho) \leq \text{bit}(n + 2F_0 + 1) + E$. Here $\text{bit}(n + 2F_0 + 1) = O(N)$ because $n \leq N$ and $\text{bit}(F_0) = O(N)$. By (16), (19), (21) and $s, m, n \leq N$, we have $E \leq 1 + \max\{\log_2 R_M, \log_2 R_D\} = (N + 1)2^{O(n)}$. This proves (14). $\square$

Comparison with the lower bound. The chain instance of Theorem 3.1 satisfies Assumption 4.1. It has $n = d + 1$ continuous variables, and its zero-multiplier threshold is $1/\delta_d = 2^{2^d}$. So every coefficient exact at multiplier zero needs at least $2^d + 1$ bits, while Theorem 4.2 gives $(N + 1)2^{O(n)}$ bits. Thus the exponential dependence on n in (14) cannot be removed. Remark 3.5 shows that the factor N is also needed: for each $n \geq 2$ and each N at least of order $n \log n$, it gives instances with n continuous variables and input length $O(N)$ whose least penalty ρ_* has $\Omega(N2^n)$ bits. Every coefficient that is exact in any of the senses of Section 2 is at least ρ_* . Hence the worst-case bit length of a sufficient coefficient is $N2^{\Theta(n)}$, for N at least of order $n \log n$. The constant in the exponent $O(n)$ is not optimized. Numerically, the coefficient may be as large as $2^{(N+1)2^{O(n)}}$. Section 5 shows that a superpolynomial bit length also requires many native inequalities that are nonlinear in the continuous variables.

5 Polynomial encoding with few nonlinear inequalities

This section proves a second upper bound: if at most k native inequalities are nonlinear in the continuous variables, some sufficient coefficient has $N^{O(k+1)}$ bits. For fixed k this is polynomial in the input length, however large the continuous dimension, the number of affine rows or the number of integer assignments. In practical terms, many linear rows together with a few convex quadratic rows cannot produce the doubly exponential growth of Section 3.

We work in the setting of Section 4. Let k be an upper bound on the number of native inequalities $g_i(x, z) \leq 0$ that are nonlinear in the continuous variables x . Constraints that are affine in x are not counted, even if they contain integer variables, such as $x_1 z_1 + z_1^2 \leq 3$; the objective is not counted either. As noted after Assumption 4.1, whether a row is nonlinear in x does not depend on z .

Theorem 5.1. *Suppose that Assumption 4.1 holds and that at most k native inequalities are nonlinear in the continuous variables. Then there is an integer $\rho \geq 1$ with*

$$\text{bit}(\rho) \leq N^{O(k+1)} \quad (23)$$

that is minimizer-exact at multiplier zero for both the infinity norm and the one-norm; the same ρ works for both. The constant in the exponent is absolute.

Neither Theorem 4.2 nor Theorem 5.1 implies the other: the first is polynomial for fixed n , the second for fixed k .

Proof strategy. In the proof of Theorem 4.2, both the separation of equality-infeasible slices and the linking multipliers of equality-feasible slices are bounded through radii of semialgebraic sets in about $2n$ variables, which costs a factor $2^{O(n)}$ in the exponent. Here each slice needs only one scalar. For an equality-infeasible slice it is the separation $\delta_z = \min_{X_z} \|r\|_\infty$. For an equality-feasible slice it is a Slater margin γ_z , the largest amount (capped at one) by which some point of the slice with $r = 0$ satisfies all nonlinear rows strictly. Both are optimal values of convex quadratic problems with at most k nonlinear rows. Lemma 5.2 below bounds such nonzero optimal values below by $2^{-N^{O(k+1)}}$; here k enters only the exponent of N , and n does not enter it. The Slater margin then bounds the multipliers of the nonlinear rows, and the linking multipliers are recovered from the remaining affine stationarity equations by linear algebra (Cramer's rule).

5.1 An algebraic bound on optimal values

The lemma says that the optimal value of a convex quadratic problem with k quadratic rows is an algebraic number whose degree and coefficient size are polynomial in the input for fixed k ; a nonzero such number cannot be smaller than $2^{-\text{poly}}$.

Lemma 5.2. *Minimize a rational convex quadratic function over a nonempty set described by a rational polyhedron with finite coordinate bounds and at most k rational convex quadratic inequalities. Let $N \geq 2$ be the input length of this problem, encoded explicitly as in Assumption 4.1, and let ω be its optimal value. Then ω is a root of a nonzero univariate integer polynomial whose degree and coefficient bitsize are at most $N^{O(k+1)}$. In particular, if $\omega \neq 0$, then*

$$|\omega| \geq 2^{-N^{O(k+1)}}. \quad (24)$$

No Slater condition or other constraint qualification is required.

Proof. The proof has three steps. (1) We delete the affine inequalities that are inactive at an optimum and substitute the active affine face; this leaves an unconstrained affine parametrization and at most $k + 1$ quadratic rows. (2) We regularize the objective and relax the rows by a parameter $\eta > 0$, so that KKT conditions hold with a positive definite matrix, and eliminate the primal variables with the adjugate; this gives polynomial conditions in η , the value w and the multipliers. (3) We express the limit $\eta \downarrow 0$ by a formula with two quantifier blocks, of sizes 1 and at most $k + 3$, and apply effective quantifier elimination, whose degree and bitsize bounds are exponential only in the block sizes.

Write the problem in variables $x \in \mathbb{R}^n$ as the minimization of $\phi_0(x)$ over the points of the polyhedron P that satisfy $\phi_i(x) \leq 0$ for $1 \leq i \leq k$. The feasible set is nonempty and compact, so an optimum x^* exists. Note that $n \leq N$.

Step 1: affine-face reduction. Let E be the affine space defined by all equalities of P and all affine inequalities of P that are active at x^* . The remaining affine inequalities of P have positive slack at x^* , hence on a neighborhood W of x^* . Deleting them leaves x^* optimal over $E \cap \{\phi_i \leq 0, 1 \leq i \leq k\}$. Indeed, suppose a point y of this set had $\phi_0(y) < \phi_0(x^*)$. By convexity, every point $x^* + \theta(y - x^*)$ with $0 < \theta \leq 1$ lies in E , satisfies the inequalities $\phi_i \leq 0$, and has objective value below $\phi_0(x^*)$. For small θ it also lies in W , so it satisfies the deleted rows and is feasible for the original problem. This contradicts the optimality of x^* .

Choose independent equations defining E . Gaussian elimination writes $E = \{x_0 + Vu : u \in \mathbb{R}^{n'}\}$, where x_0 and V are rational with bit length polynomial in N , and the free coordinates u are n' of the coordinates of x ; thus $n' \leq n \leq N$. If $n' = 0$, then $E = \{x^*\}$ with $x^* = x_0$, so $\omega = \phi_0(x_0)$ is rational with bit length polynomial in N ; it is the root of a linear integer polynomial with such coefficients, and the lemma follows. Assume $n' \geq 1$. Let $R_{\text{box}} \geq 1$ be an integer of bit length polynomial in N that bounds the absolute values of all coordinates on P . Since u^* is a subvector of x^* , we have $\|u^*\|_2^2 \leq n' R_{\text{box}}^2$. Add the ball constraint $\|u\|_2^2 - n' R_{\text{box}}^2 - 1 \leq 0$. It holds strictly at u^* , keeps the optimal value ω attained at u^* , and makes the feasible set compact.

We now have an objective $\psi_0(u) = \phi_0(x_0 + Vu)$ and $k' \leq k + 1$ convex quadratic inequalities $\psi_i(u) \leq 0$: the substituted rows $\phi_i(x_0 + Vu) \leq 0$ and the ball. There are no remaining affine constraints, and all coefficients have bitsize polynomial in N . Some ψ_i may be affine or constant; this does not affect the argument. Write

$$\psi_i(u) = \frac{1}{2}u^T Q_i u + a_i^T u + c_i \quad (0 \leq i \leq k'),$$

with symmetric $Q_i \succeq 0$. The matrices Q_i may be singular.

Step 2: regularization and elimination. For $0 < \eta < 1$, define

$$\omega(\eta) = \min\{\psi_0(u) + \eta\|u\|_2^2 : \psi_i(u) \leq \eta \ (1 \leq i \leq k')\}. \quad (25)$$

The relaxed ball bounds every feasible point, so the feasible set is compact, and it contains u^* ; hence an optimum exists. The point u^* satisfies every relaxed inequality strictly, so it is a Slater point of this convex problem, and nonnegative KKT multipliers μ_i exist at every optimum. (Strict convexity of the objective only adds uniqueness of the optimum, which we do not need.) Stationarity reads

$$H(\eta, \mu) u = -a_0 - \sum_{i=1}^{k'} \mu_i a_i, \quad H(\eta, \mu) = Q_0 + 2\eta I + \sum_{i=1}^{k'} \mu_i Q_i. \quad (26)$$

The term $2\eta I$ makes $H(\eta, \mu)$ positive definite for all $\eta > 0$ and all $\mu \geq 0$, even when every Q_i is singular. This allows us to solve for u . Let

$$\Delta(\eta, \mu) = \det H(\eta, \mu), \quad U(\eta, \mu) = -\text{adj}(H(\eta, \mu)) \left(a_0 + \sum_{i=1}^{k'} \mu_i a_i \right).$$

The entries of $H(\eta, \mu)$ are affine in (η, μ) , so Δ and the entries of U are polynomials of degree at most n' , and $u = U/\Delta$. We abbreviate $\Delta = \Delta(\eta, \mu)$ and $U = U(\eta, \mu)$ and define

$$\Phi_i = \frac{1}{2}U^T Q_i U + \Delta a_i^T U + (c_i - \eta)\Delta^2 \quad (1 \leq i \leq k'), \quad (27)$$

$$\Phi_0 = \frac{1}{2}U^T Q_0 U + \Delta a_0^T U + (c_0 - w)\Delta^2 + \eta U^T U. \quad (28)$$

For $\Delta \neq 0$ these are $\Phi_i = \Delta^2(\psi_i(U/\Delta) - \eta)$ and $\Phi_0 = \Delta^2(\psi_0(U/\Delta) + \eta\|U/\Delta\|_2^2 - w)$. Consider the polynomial system

$$\Sigma(\eta, w, \mu) : \quad \Delta > 0, \quad \mu_i \geq 0, \quad \Phi_i \leq 0, \quad \mu_i \Phi_i = 0 \quad (1 \leq i \leq k'), \quad \Phi_0 = 0. \quad (29)$$

The condition $\Delta > 0$ is implied by $\eta > 0$ and $\mu \geq 0$; we keep it for clarity. For $0 < \eta < 1$, the system $\Sigma(\eta, w, \mu)$ has a solution μ exactly when $w = \omega(\eta)$. If $w = \omega(\eta)$, an optimum and its KKT multipliers give a solution: stationarity gives $u = U/\Delta$, and feasibility, complementarity and the optimal value give the remaining conditions. Conversely, if μ solves the system, then $u = U/\Delta$ satisfies stationarity, feasibility and complementarity, so (u, μ) is a KKT pair of a convex problem; hence u is optimal and $\Phi_0 = 0$ gives $w = \omega(\eta)$.

Every polynomial in (29) has degree at most $2n' + 2$. Its coefficients have bitsize polynomial in N . To see this, clear the denominators of the entries of the matrices Q_i by one positive common denominator. Each coefficient of the determinant, as a polynomial in (η, μ) , is then a sum of at most $n'!(k' + 2)^{n'}$ products of at most n' integers of polynomial bitsize; since $n' \leq N$ and $k' \leq N + 1$, its bitsize is polynomial in N . The adjugate and the polynomials Φ_i have the same property. This bounds coefficient heights even when the number of monomials is large. Multiplying each polynomial by a positive integer that clears its remaining denominators does not change the sign conditions.

Step 3: the limit and its algebraic height. We have $\omega(\eta) \rightarrow \omega$ as $\eta \downarrow 0$. For the upper limit, u^* is feasible for (25), so $\omega(\eta) \leq \omega + \eta\|u^*\|_2^2$. For the lower limit, take any sequence $\eta_j \downarrow 0$ and regularized optima u_j . They lie in a common ball, so a subsequence converges to some $\bar{u}$. The limit satisfies $\psi_i(\bar{u}) \leq 0$ for all i , so $\psi_0(\bar{u}) \geq \omega$, and $\omega(\eta_j) \geq \psi_0(u_j)$ gives $\liminf_j \omega(\eta_j) \geq \omega$ along the subsequence. Applying this to a subsequence that realizes the $\liminf$ proves the claim. No bound on the regularized multipliers is needed.

Consequently the following formula, with one free variable α , defines exactly the set $\{\omega\}$:

$$\forall \xi \exists (\eta, w, \mu) \left[\xi \leq 0 \vee \left(0 < \eta < 1, \quad \eta < \xi, \quad -\xi < w - \alpha < \xi, \quad \Sigma(\eta, w, \mu) \right) \right]. \quad (30)$$

Since $\xi \leq 0$ does not involve η, w or μ , this prenex formula is equivalent to $\forall \xi [\xi \leq 0 \vee \exists (\eta, w, \mu) (\dots)]$. For $\alpha = \omega$, convergence gives, for every $\xi > 0$, some $0 < \eta < \min\{1, \xi\}$ with $|\omega(\eta) - \omega| < \xi$, and $w = \omega(\eta)$ with KKT multipliers μ is a witness. Conversely, if the formula holds for α , then witnesses for $\xi = 1/j$ give $\eta_j < 1/j$ and $w_j = \omega(\eta_j)$ with $|w_j - \alpha| < 1/j$; since $\omega(\eta_j) \rightarrow \omega$, we get $\alpha = \omega$.

We apply effective block quantifier elimination [6, Theorem 14.16], also stated in Basu [4, Theorem 2.27]. For a prenex formula with quantifier blocks of sizes b_1 and b_2 , ℓ free variables, polynomials of degree at most D , and integer coefficients of bitsize at most τ , it gives an equivalent quantifier-free formula whose polynomials have degree at most $D^{O(b_1)O(b_2)}$ and integer coefficients of bitsize at most $\tau D^{O(b_1)O(b_2)O(\ell)}$. We use this bound on coefficient heights, not only the bound on the number of arithmetic operations. For (30) the parameters are

$$b_1 = 1 \text{ (the variable } \xi), \quad b_2 = k' + 2 \leq k + 3 \text{ (the variables } \eta, w, \mu), \quad \ell = 1 \text{ (the variable } \alpha),$$

$$D = 2n' + 2 = O(n' + 1), \quad \tau = N^{O(1)}.$$

Since $n' \leq N$, we have $D^{O(k+1)} = N^{O(k+1)}$. The quantifier-free formula therefore involves univariate integer polynomials in α of degree and coefficient bitsize $N^{O(k+1)}$.

Remove the identically zero polynomials from this formula; their signs are constant. At least one remaining polynomial vanishes at ω . Otherwise every remaining polynomial would have a constant nonzero sign on a neighborhood of ω , so the formula would have a constant truth value there, whereas it defines the singleton $\{\omega\}$. This proves the claim on degree and height.

Now let $\omega \neq 0$. Divide that polynomial by the largest power of the indeterminate that divides it. The quotient still vanishes at ω , has a nonzero constant term, and its coefficients are among those of the original polynomial; let τ_1 bound their bitsize. The reciprocal polynomial has the root $1/\omega$ and a leading coefficient of absolute value at least one, so the Cauchy bound gives $|1/\omega| \leq 1 + 2^{\tau_1}$, that is, $|\omega| \geq 1/(1 + 2^{\tau_1})$. Absorbing the additive constant in the exponent proves (24). $\square$

The argument neither finds the active affine face algorithmically nor needs a genericity assumption. Algebraic methods for few quadratic constraints have direct precedents. Grigoriev and Pasechnik [21, Theorem 1.2] give sampling and coefficient bounds for zero sets of polynomials in quadratic maps; their Theorem 1.5 states an optimization version, whose proof is deferred to a continuation of their paper. Kamminga and Rudolph [26, Sections 6.2–7.2 and 8.5] give reduced-variable formulas for optimization under a boundedness hypothesis on the solution set, and an application to QCQP with a fixed total number of constraints. After our affine-face reduction, squared slack variables would turn the inequalities $\psi_i \leq 0$ into a bounded zero set of a quadratic map (bounded because u lies in the ball and each squared slack equals $-\psi_i(u)$, which is bounded there), which connects the lemma to these results. We use instead a direct convex KKT elimination and the explicit two-block limit formula (30). The affine-face reduction is what allows arbitrarily many affine rows. The value bound itself follows from established algebraic tools; the new part is its application to exact penalties, with the explicit dependence on the number of nonlinear rows.

5.2 Separation and multiplier bounds

Proof of Theorem 5.1. We bound the separation of every equality-infeasible slice from below and a linking multiplier of every equality-feasible slice from above, each by applying Lemma 5.2 to an auxiliary problem, and then apply Lemma 2.6. Use the objective bound $|f| \leq F_0$ from Section 4. Every auxiliary problem below is obtained by substituting a boxed integer assignment z . Its coefficients keep bitsize $O(N)$, as in (16), so its explicit input length is at most N^c for an absolute constant c , uniformly in z . Hence the bound $2^{-(N^c)^{O(k+1)}}$ of Lemma 5.2 is again $2^{-N^{O(k+1)}}$.

Equality-infeasible slices. Let X_z be a nonempty equality-infeasible slice. Let R_0 be a rational bound on all residual magnitudes $|r_j(x, z)|$ over the input box, with bit length polynomial in N . Minimize t over the points (x, t) with $x \in X_z$ and

$$-t \leq r_j(x, z) \leq t \quad (1 \leq j \leq m), \quad 0 \leq t \leq R_0.$$

This is a nonempty compact convex quadratic problem whose only nonlinear rows are the at most k native rows that are nonlinear in x . Its optimal value is $\delta_z = \min_{X_z} \|r\|_\infty$, which is positive by compactness. Lemma 5.2 gives

$$\delta_z \geq 2^{-N^{O(k+1)}}. \quad (31)$$

Equality-feasible slices: the Slater margin. Let X_z be an equality-feasible slice with at least one row nonlinear in x . Let P_z be the polyhedron defined by the native constraints that are affine in x (including the bounds) and by the relaxed equalities $r(x, z) = 0$. Define the common Slater margin of the nonlinear rows,

$$\gamma_z = \max\{t : x \in P_z, \ g_i(x, z) + t \leq 0 \text{ for all rows } i \text{ nonlinear in } x, \ 0 \leq t \leq 1\}. \quad (32)$$

The refined Slater condition provides a point of P_z at which every nonlinear row is strict, so $\gamma_z > 0$. Minimizing $-t$ in this compact convex problem with at most k nonlinear rows and applying Lemma 5.2 gives

$$\gamma_z \geq 2^{-N^{O(k+1)}}. \quad (33)$$

Multipliers of the nonlinear rows. Let $(\bar{x}, \gamma_z)$ be optimal in (32), and let x^* be an optimum of the slice problem $v_z = \min\{f(x, z) : x \in X_z, r(x, z) = 0\}$. By the refined Slater condition, x^* has a KKT certificate, as in Section 4, with multipliers $\mu_i \geq 0$ for the nonlinear rows, nonnegative multipliers for the native rows affine in x , and linking multipliers for $r = 0$. The corresponding convex Lagrangian is minimized at x^* , where its value is v_z . Evaluate it at $\bar{x}$: the linking terms vanish because $r(\bar{x}, z) = 0$, the terms of the affine native rows are nonpositive because $\bar{x} \in P_z$, and each nonlinear row contributes at most $-\gamma_z \mu_i$. Hence

$$v_z \leq f(\bar{x}, z) - \gamma_z \sum_i \mu_i, \quad \sum_i \mu_i \leq \frac{2F_0}{\gamma_z}. \quad (34)$$

This is the usual bound on multipliers from a Slater point with a margin.

Linking multipliers by Cramer's rule. It remains to bound the linking multipliers; we do not need to bound the multipliers of the affine native rows in this particular certificate. The vector

$$b_* = -\nabla_x f(x^*, z) - \sum_i \mu_i \nabla_x g_i(x^*, z)$$

has coordinates bounded by $2^{N^{O(1)}}(1 + 2F_0/\gamma_z)$, since all gradients are affine with coefficients of polynomial bitsize and x^* and z lie in their boxes. By (33), this bound is $2^{N^{O(k+1)}}$. By stationarity, b_* lies in the cone generated by the gradients of the active native rows affine in x and by both signs of the linking normals A_j^T . By the conic Carathéodory theorem, b_* is a nonnegative combination of linearly independent generators from this set; let $\nu \leq n$ be their number. Their $n \times \nu$ column matrix has rank ν , so some ν of its rows form a nonsingular square submatrix. Its entries are rational input coefficients after the integer substitution, so determinant bounds and Cramer's rule bound the entries of its inverse by $2^{N^{O(1)}}$. Applying the inverse to the corresponding ν coordinates of b_* bounds all selected coefficients by $2^{N^{O(k+1)}}$. This bound holds for real coefficients; the primal point x^* and the multipliers μ_i need not be rational.

Combining the coefficients of oppositely signed linking normals gives a new KKT certificate at x^* : it keeps the multipliers μ_i , puts the selected coefficients on the active affine rows, and has a linking multiplier λ_z with

$$\|\lambda_z\|_1 \leq 2^{N^{O(k+1)}}. \quad (35)$$

Only active affine rows received multipliers and the multipliers μ_i were kept, so stationarity and complementarity still hold. As in Section 4, the new Lagrangian is convex and minimized at x^* with value v_z ; dropping its nonpositive native terms on X_z gives $f(x, z) + \lambda_z^T r(x, z) \geq v_z$ for all $x \in X_z$. If the slice has no row nonlinear in x , the same argument applies directly with $b_* = -\nabla_x f(x^*, z)$; KKT conditions hold without a constraint qualification for affine constraints, and no margin problem is needed.

Assembly. All constants are uniform over the integer assignments. Choose an integer ρ strictly larger than the bound in (35) and than $2F_0$ times the reciprocal of the bound in (31). Its bit length is $N^{O(k+1)}$. The dual norm of $\|\cdot\|_\infty$ is $\|\cdot\|_1$, so Lemma 2.6, with objective bounds $\underline{f} = -F_0$ and $\bar{f} = F_0$, shows that ρ is minimizer-exact at multiplier zero for the infinity norm. The same ρ works for the one-norm, because $\|r\|_1 \geq \|r\|_\infty$ and both norms vanish on the original feasible set. $\square$

Remark 5.3. By Theorem 5.1, superpolynomial penalty encoding within this model needs a growing number of nonlinear inequalities; arbitrarily many affine rows alone do not produce it. Some growth with k is unavoidable: the chain of Remark 3.5 has $k = d - 1$ and needs $\Omega(N2^k)$ bits, against the $N^{O(k+1)}$ bits of Theorem 5.1.

5.3 A conservative coefficient can be printed in polynomial time

Because the encoding bounds of Theorems 4.2 and 5.1 are uniform, a sufficient coefficient can be output without finding an optimum, a Slater point or an active face.

Corollary 5.4. *For each fixed continuous dimension n , and separately for each fixed $k \geq 0$, there is a polynomial-time algorithm with the following property. On every input that is promised to satisfy Assumption 4.1 (and, in the second case, to have at most k native inequalities nonlinear in the continuous variables), it outputs in binary an integer ρ that is minimizer-exact at multiplier zero for both the infinity norm and the one-norm.*

Proof. Choose a universal integer constant C such that the bounds in Theorems 4.2 and 5.1 are at most

$$B_{\text{dim}} = (N + 1)2^{C(n+1)} \quad \text{and} \quad B_{\text{cnt}} = N^{C(k+1)},$$

respectively. In each case some integer that is minimizer-exact at multiplier zero has at most B_{dim} , respectively B_{cnt} , bits and is therefore smaller than $2^{B_{\text{dim}}}$, respectively $2^{B_{\text{cnt}}}$. By the consequences of (2), every larger coefficient is minimizer-exact at multiplier zero as well. The algorithm therefore prints a one followed by B_{dim} , respectively B_{cnt} , zeros. For fixed n , respectively fixed k , this output and the time to produce it are polynomial in N . $\square$

For native constraints linear in all variables, a special case of $k = 0$, the analogous results are known: polynomial encoding length for a jointly convex quadratic objective [22, Theorem 11], and polynomial-time output for MILPs with integer data [29, Theorem 15], whose Table 1 remarks that the extension to convex MIQPs is easy (Section 1.2). The case $k = 0$ of Corollary 5.4 also covers rows that are affine in x but contain products with integer variables, such as $x_1 z_1 + z_1^2 \leq 3$. Because all variables are bounded, such rows can also be rewritten exactly as linear rows of polynomial size, through a binary expansion of z and exact linearization of the resulting products, after which Gu et al. [22, Theorem 11] applies when the objective is jointly convex; this part of the corollary is therefore mainly a convenience. The output of Corollary 5.4 is a conservative and numerically huge coefficient: the algorithm does not verify the promises, solve the problem or approximate the least penalty, and the constant C exists by the cited effective bounds but has not been evaluated.

6 Hardness of calibrating the least penalty

This section shows that a sufficient coefficient can be trivial to write down while a nearly least one is hard to compute. On a binary box with one relaxed equality, a linear objective, a known unique optimizer and the known sufficient coefficient $\rho = 1$, no polynomial-time algorithm approximates the least penalty within any polynomial factor unless $P = NP$; the same holds for the zero-multiplier threshold. The hardness needs a growing number of binary variables; fixed dimension is not covered.

Throughout, $m = 1$ and the norm is the absolute value. On $\mathbb{R}$ every norm is a positive multiple of $|\cdot|$, which only rescales the thresholds (Remark 6.7).

6.1 The calibration problem

Definition 6.1 (Calibration). Fix a family of instances of (1) with rational data, encoded in binary as in Section 2, and a function φ from the positive integers to $[1, \infty)$. Given an instance of encoding length N whose least penalty ρ_* is finite, *sufficient calibration within factor φ* asks for a rational $\hat{\rho}$ with

$$\rho_* \leq \hat{\rho} \leq \varphi(N) \rho_*, \quad (36)$$

and *two-sided calibration within factor φ* asks for a rational $\hat{\rho}$ with

$$\rho_*/\varphi(N) \leq \hat{\rho} \leq \varphi(N) \rho_*. \quad (37)$$

Calibration at multiplier zero is defined in the same way, with $\rho_*(0)$ in place of ρ_* . *Polynomial-factor calibration* means that $\varphi = p$ is a fixed polynomial with $p(N) \geq 1$ for all N .

A one-sided estimate (36) is itself a sufficient coefficient: the value-exact coefficients form the interval $[\rho_*, \infty)$, and the coefficients exact at multiplier zero form $[\rho_*(0), \infty)$ (Corollary 2.3 and the discussion after it). A two-sided estimate need not be sufficient. On the families below a sufficient coefficient is known in advance, so the only question is how close to the least one an algorithm can get.

6.2 A binary-box family

Take a SUBSET SUM instance: positive integers $w_1, \dots, w_t$ and a positive integer target β , with encoding length N_0 . For $I \subseteq \{1, \dots, t\}$ write $w(I) = \sum_{i \in I} w_i$. The instance is a YES instance if $w(I) = \beta$ for some I , and a NO instance otherwise. Deciding this is NP-complete. Append the item $w_{t+1} = \beta + 1$, extend the notation $w(I)$ to $I \subseteq \{1, \dots, t+1\}$, choose an integer $\kappa \geq 2$, and set

$$X = \{0, 1\}^{t+2} \ni (x, q), \quad f(x, q) = -q, \quad r(x, q) = \kappa \sum_{i=1}^{t+1} w_i x_i - (\kappa\beta + 1)q. \quad (38)$$

The instance is given by the integer coefficient vector $(\kappa w_1, \dots, \kappa w_{t+1}, -(\kappa\beta + 1))$ of r ; the box and the objective are implicit.

How the gadget works. Read x as the indicator vector of a subset I and q as a switch whose slice $q = 1$ is improving: it lowers the objective from 0 to -1 . The residual is $\kappa w(I) \geq 0$ at $q = 0$ and $\kappa(w(I) - \beta) - 1$ at $q = 1$. By Proposition 2.4, the least penalty is governed by how close the improving slice $q = 1$ comes to $r = 0$ from each side. Each part of the construction has one purpose.

- The $+1$ in $\kappa\beta + 1$ makes the whole slice $q = 1$ equality-infeasible: $r = 0$ at $q = 1$ would require κ to divide $\kappa\beta + 1$. At $q = 0$, $r = 0$ forces $x = 0$ because all items are positive. Hence $(x, q) = (0, 0)$ is the unique feasible point and $v = 0$, known without solving SUBSET SUM.
- The appended item $\beta + 1$ fixes the nearest positive residual at $q = 1$ to $\kappa - 1$ on every instance, so the positive side carries no information about the instance.
- The factor κ separates the cases on the negative side: a subset with $w(I) = \beta$ gives residual -1 , while every subset with $w(I) \leq \beta - 1$ gives residual at most $-(\kappa + 1)$.

Linear optimization over X is trivial; this does not make the penalized subproblem with the absolute residual easy.

Formally, let $\beta_- = \max\{w(I) : I \subseteq \{1, \dots, t+1\}, w(I) \leq \beta\}$ be the largest attainable sum not exceeding the target (the empty set shows that it exists), and define

$$g_- = \kappa(\beta - \beta_-) + 1, \quad g_+ = \kappa - 1. \quad (39)$$

At $q = 1$ the residual $\kappa(w(I) - \beta) - 1$ is negative exactly when $w(I) \leq \beta$; its magnitude is then $\kappa(\beta - w(I)) + 1 \geq g_-$, with equality when $w(I) = \beta_-$. The residual is positive exactly when $w(I) \geq \beta + 1$; it is then at least $\kappa - 1 = g_+$, with equality for $I = \{t+1\}$. No subset with sum at most β contains the item $t+1$, so β_- is the best sum of the original instance. Consequently

$$g_- = 1 \text{ in YES instances,} \quad g_- \geq \kappa + 1 \text{ in NO instances.} \quad (40)$$

Proposition 6.2. *Let $w_1, \dots, w_t, \beta$ be positive integers, $w_{t+1} = \beta + 1$, and $\kappa \geq 2$ an integer. The instance (38) has $v = 0$, attained only at $(x, q) = (0, 0)$. For every $\rho \geq 0$,*

$$D_\rho = \min \left\{ 0, -1 + \rho \frac{2g_- - g_+}{g_- + g_+} \right\}, \quad (41)$$

and the multiplier $\lambda_\rho = \rho(g_- - g_+)/(g_- + g_+)$ attains this value. Consequently

$$\rho_* = \frac{1}{2} \left(\frac{1}{g_-} + \frac{1}{g_+} \right), \quad \rho_*(0) = \max \left\{ \frac{1}{g_-}, \frac{1}{g_+} \right\}. \quad (42)$$

Proof. The plan is to bound D_ρ above by a two-point balanced mixture and to show that λ_ρ attains this bound. The value $v = 0$ and the uniqueness of the feasible point were shown above.

Let P_- be a point with $q = 1$ and $w(I) = \beta_-$, and let P_+ be the point with $q = 1$ and $I = \{t+1\}$. Then $(f, r) = (-1, -g_-)$ at P_- and $(f, r) = (-1, g_+)$ at P_+ . The weights $g_+/(g_- + g_+)$ on P_- and $g_-/(g_- + g_+)$ on P_+ give a balanced mixture: its average residual is zero. Its average of $f + \rho|r|$ is $-1 + 2\rho g_- g_+/(g_- + g_+)$, so Proposition 2.2 bounds D_ρ by this number. Weak duality bounds D_ρ by $v = 0$. This proves “ $\leq$ ” in (41).

For the reverse inequality, note that $\rho - \lambda_\rho = 2\rho g_+/(g_- + g_+) \geq 0$ and $\rho + \lambda_\rho = 2\rho g_-/(g_- + g_+) \geq 0$. A native point with $q = 1$ and $r < 0$ has $|r| \geq g_-$, so its penalized value at λ_ρ is

$$-1 + (\rho - \lambda_\rho)|r| \geq -1 + (\rho - \lambda_\rho)g_- = -1 + \rho \frac{2g_- - g_+}{g_- + g_+}.$$

A native point with $q = 1$ and $r > 0$ has $r \geq g_+$, and $-1 + (\rho + \lambda_\rho)r \geq -1 + (\rho + \lambda_\rho)g_+$ is the same number. A native point with $q = 0$ has $f = 0$ and $r \geq 0$, so its value $(\rho + \lambda_\rho)r$ is nonnegative. Hence $L_\rho(\lambda_\rho)$ is at least the right-hand side of (41), and the upper bound is attained.

By (41), $D_\rho = 0$ exactly when $\rho \geq (g_- + g_+)/(2g_- - g_+)$, which is the formula for ρ_* . For $\rho_*(0)$, apply (2) with $\lambda = 0$ and $v = 0$: an infeasible point with $q = 0$ contributes $0/|r| = 0$, and an infeasible point with $q = 1$ contributes $1/|r|$, which is largest at the smallest residual magnitudes g_- and g_+ . $\square$

The formulas agree with Proposition 2.4, where $A_+ = 1/g_+$ (points with $q = 0$ contribute 0) and $A_- = 1/g_-$. At $\rho = \rho_*$ the interval of exact multipliers in (5) reduces to the single multiplier $\lambda_{\rho_*} = (1/g_+ - 1/g_-)/2$, and both witnesses P_- and P_+ tie with the optimum. So ρ_* is exact at λ_{ρ_*} but not minimizer-exact. Every $\rho > \rho_*$ is minimizer-exact at λ_{ρ_*} : then $\rho - \lambda_{\rho_*} > 1/g_-$ and $\rho + \lambda_{\rho_*} > 1/g_+$, so every infeasible point has a positive penalized value. On this family $1/(2\kappa) < \rho_* \leq 1$, because $g_+ = \kappa - 1$ and $g_-, g_+ \geq 1$.

A sufficient coefficient is known. For every instance of the family, $\rho = 1$ is exact at multiplier zero. Every nonzero residual is an integer and the objective is 0 or -1 , so $f + |r| \geq 0$ at every native point. An infeasible point has $f + |r| = 0$ only if $q = 1$ and $|r| = 1$. The residual -1 occurs exactly in YES instances, and the residual $+1$ occurs exactly when $\kappa = 2$ (through $I = \{t + 1\}$). These ties are the only obstruction: every $\rho > 1$ is minimizer-exact at multiplier zero.

The YES/NO gap. By (40) and (42),

$$\begin{aligned} \rho_* &= \frac{\kappa}{2(\kappa - 1)} > \frac{1}{2}, & \rho_*(0) &= 1 & \text{in YES instances,} \\ 0 < \rho_* &\leq \frac{\kappa}{\kappa^2 - 1}, & \rho_*(0) &= \frac{1}{\kappa - 1} & \text{in NO instances.} \end{aligned} \tag{43}$$

The ratio between the YES value and the NO bound is $(\kappa + 1)/2$ for ρ_* and $\kappa - 1$ for $\rho_*(0)$. For $\kappa = 2$ the zero-multiplier threshold equals 1 in both cases, so part (b) below uses $\kappa \geq 3$.

6.3 Testing exactness and calibrating are hard

The first result shows that testing a single given coefficient is already hard. The membership argument uses that $v = 0$ is known on this family; for a general instance, certifying $D_\rho < v$ would also require information about v .

Theorem 6.3. *Consider the instances (38) with $\kappa = 4$, each given by the integer coefficient vector $(4w_1, \dots, 4w_{t+1}, -(4\beta + 1))$ of r . Deciding whether $\rho = 1/3$ is value-exact, that is, whether $D_{1/3} = 0$, is coNP-complete. More precisely, $D_{1/3} = -1/2$ for YES instances of SUBSET SUM and $D_{1/3} = 0$ for NO instances.*

Proof. For YES instances, $g_- = 1$ and $g_+ = 3$, and (41) gives $D_{1/3} = \min\{0, -1 + \frac{1}{3} \cdot \frac{6}{4}\} = -1/2$. For NO instances, $\rho_* \leq 4/15 < 1/3$ by (43), so $D_{1/3} = 0$.

Hardness: the map from a SUBSET SUM instance to the coefficient vector is computable in polynomial time and sends NO instances to exact coefficients and YES instances to non-exact ones. It is therefore a polynomial reduction from the complement of SUBSET SUM, which is coNP-complete.

Membership in coNP: whether an input vector belongs to the family can be checked in polynomial time, since β , the items w_i and the condition $w_{t+1} = \beta + 1$ are recovered by integer division by 4. On a member of the family, $D_{1/3} < 0$ holds exactly in YES instances, and a subset $I \subseteq \{1, \dots, t\}$ with $w(I) = \beta$ is a polynomial certificate of this. $\square$

Consequently, computing ρ_* on this family is NP-hard, and evaluating $D_{1/3}$ within absolute error strictly below $1/4$ decides SUBSET SUM. In contrast, $D_0 = -1$ on every instance: without the norm term the Lagrangian dual is trivial to evaluate and never closes the gap.

The second result says that even approximating the threshold, from above, within a polynomial factor is hard.

Theorem 6.4. *Let p be a fixed polynomial with $p(N) \geq 1$ for all N . Unless $P = NP$, no polynomial-time algorithm achieves sufficient calibration within factor p on the family (38), either*

(a) *for the least penalty: on every instance of encoding length N , return a rational $\hat{\rho}$ with*

$$\rho_* \leq \hat{\rho} \leq p(N) \rho_*; \tag{44}$$

(b) or at multiplier zero: on every instance of encoding length N , return a rational $\hat{\rho}$ with $\rho_*(0) \leq \hat{\rho} \leq p(N) \rho_*(0)$, where $\rho_*(0) = \max\{1/g_-, 1/g_+\}$; this equals 1 in YES instances and $1/(\kappa - 1)$ in NO instances.

Proof. The plan is to choose κ exponentially large, so that the YES/NO ratio in (43) exceeds $p(N)$, and then to read off the SUBSET SUM answer by comparing $\hat{\rho}$ with a fixed number.

Given a SUBSET SUM instance of length N_0 , take $\kappa = 2^{N_0}$. The constructed instance has $t + 2 \leq N_0 + 2$ coefficients, each with $O(N_0)$ bits, so its length is $N = O(N_0^2)$ and it is computed in polynomial time. In particular $p(N)$ is bounded by a polynomial in N_0 .

(a) In YES instances every admissible output satisfies $\hat{\rho} \geq \rho_* > 1/2$. In NO instances every admissible output satisfies

$$\hat{\rho} \leq p(N) \frac{\kappa}{\kappa^2 - 1} = p(N) \frac{2^{N_0}}{4^{N_0} - 1} < \frac{1}{2}$$

for all sufficiently large N_0 , since the exponential term dominates. (b) For $N_0 \geq 2$ we have $\kappa \geq 3$. In YES instances every admissible output satisfies $\hat{\rho} \geq 1$; in NO instances $\hat{\rho} \leq p(N)/(2^{N_0} - 1) < 1$ for all sufficiently large N_0 .

In both cases there is a constant n_1 , depending only on p , such that the comparison of $\hat{\rho}$ with $1/2$ (respectively 1) decides every SUBSET SUM instance with $N_0 \geq n_1$. Instances with $N_0 < n_1$ are finitely many and are decided by exhaustive enumeration. The output $\hat{\rho}$ of a polynomial-time algorithm has polynomially many bits, so the comparison takes polynomial time. This would decide SUBSET SUM in polynomial time. $\square$

Remark 6.5 (Two-sided estimates and constant factors). Two-sided calibration is not easier. If $\hat{\rho}$ satisfies (37) with $\varphi = p$, replace p by a pointwise larger polynomial p' with positive integer coefficients (so that $p'(N)$ is an integer computable in polynomial time); then $p'(N)\hat{\rho}$ satisfies (36) with $\varphi = p'^2$, again a polynomial. Thus Theorem 6.4 also rules out two-sided polynomial-factor calibration, in both parts. For a fixed constant factor $C \geq 1$, an exponential κ is not needed. In part (a), a one-sided estimate within factor C separates the YES value $\kappa/(2(\kappa - 1))$ from the NO bound $C\kappa/(\kappa^2 - 1)$ exactly when $C < (\kappa + 1)/2$, so any constant κ with $\kappa + 1 > 2C$ suffices. In part (b), any constant κ with $\kappa - 1 > C$ suffices.

Remark 6.6 (Subexponential factors). The proof gives more than polynomial factors. For every fixed $\epsilon \in (0, 1)$, sufficient calibration within factor $\varphi(N) = 2^{N^{1-\epsilon}}$ is also impossible in polynomial time unless $P = NP$, in both parts. Choose an integer $c > (1 - \epsilon)/\epsilon$ and take $\kappa = 2^{N_0^c}$. Each coefficient then has at most $N_0^c + O(N_0)$ bits, so $N = O(N_0^{c+1})$ and the construction is polynomial. Since $(c + 1)(1 - \epsilon) < c$, we have $N^{1-\epsilon} \leq N_0^c - 2$, that is $\varphi(N) \leq \kappa/4$, for all sufficiently large N_0 . Then in part (a) every NO output satisfies $\hat{\rho} \leq \varphi(N)\kappa/(\kappa^2 - 1) \leq \kappa^2/(4(\kappa^2 - 1)) < 1/2$, while every YES output exceeds $1/2$. In part (b) every NO output satisfies $\hat{\rho} \leq \varphi(N)/(\kappa - 1) \leq \kappa/(4(\kappa - 1)) < 1$, while every YES output is at least 1.

Remark 6.7 (Rescaling does not help). Replacing r by cr for a rational $c > 0$ replaces $L_\rho(\lambda)$ by $L_{c\rho}(c\lambda)$ of the original instance; replacing the absolute value by the norm $c|\cdot|$ replaces it by $L_{c\rho}(\lambda)$. In both cases ρ_* and $\rho_*(0)$ are divided by c , and the YES/NO ratios $(\kappa + 1)/2$ and $\kappa - 1$ are unchanged. A guarantee within a factor is invariant under such rescaling, so no choice of scaling makes calibration easier on this family.

Remark 6.8 (Weak NP-hardness). The reduction is from SUBSET SUM, which is NP-hard only when its numbers are large, and the constructed coefficients are large as well. This is unavoidable on this family: a dynamic program over the attainable values of $\sum_i w_i x_i$ computes g_- , hence ρ_* and $\rho_*(0)$

exactly, in time polynomial in N and the largest coefficient magnitude. So the family gives weak NP-hardness only. Proposition 6.10 below gives strong NP-hardness of exact computation with coefficients in $\{-1, 0, 1\}$, at the price of a native set over which linear optimization is itself hard.

Remark 6.9 (Where the hardness lies). The difficulty is testing whether a given coefficient is exact (Theorem 6.3), not locating the threshold once such a test is available. Suppose an oracle decides, for a given instance of the family and a rational ρ , whether ρ is value-exact. Since $\rho_* \in (1/(2\kappa), 1]$ and the value-exact coefficients form $[\rho_*, \infty)$, testing $\rho = 2^{-j}$ for $j = 0, 1, \dots, \lceil \log_2 \kappa \rceil + 1$ finds the largest j with 2^{-j} exact; then $2^{-j} \geq \rho_* > 2^{-j-1}$, so $\hat{\rho} = 2^{-j}$ is a sufficient calibration within factor two, after $O(\log \kappa) = O(N)$ oracle calls. The same works at multiplier zero with an oracle for $L_\rho(0) = v$. In decomposition methods the penalized subproblem is solved anyway. The results above say that nearly least penalties are as hard to certify as those subproblems, even when the original problem is trivial.

6.4 Unit coefficients with a hard native set

The binary-box family has large numbers. The next family has all coefficients in $\{-1, 0, 1\}$, and its least penalty is the stability number of a graph. This gives strong NP-hardness of exact computation, because the native set now encodes a hard optimization problem.

For a simple graph $G = (V, E)$ with $V \neq \emptyset$, let $\alpha(G)$ be the maximum size of a stable set. Take binary variables x_i ($i \in V$), y_+ and y_- , with native constraints

$$y_+ + y_- \leq 1, \quad x_i \leq y_+ + y_- \quad (i \in V), \quad x_i + x_j \leq 1 \quad (ij \in E), \quad (45)$$

objective $f = -\sum_{i \in V} x_i$ and relaxed equality $r = y_+ - y_- = 0$.

Proposition 6.10. *For the instance (45), $v = 0$ is attained only at the zero vector, and for all $\rho \geq 0$ and $\lambda \in \mathbb{R}$,*

$$L_\rho(\lambda) = \min\{0, \rho - \alpha(G) - |\lambda|\}, \quad D_\rho = \min\{0, \rho - \alpha(G)\}, \quad \rho_* = \rho_*(0) = \alpha(G). \quad (46)$$

At $\rho = \alpha(G)$, $\lambda = 0$ is the only exact multiplier and the infeasible points given by maximum stable sets tie with the optimum. At multiplier zero, ρ is minimizer-exact if and only if $\rho > \alpha(G)$. Consequently, computing ρ_ is strongly NP-hard. Given G and an integer c as input, deciding whether $\rho = c$ is value-exact, that is, whether $\alpha(G) \leq c$, is coNP-complete; a stable set of size $c + 1$ certifies failure. For each fixed c , enumerating the $(c + 1)$ -subsets of V decides it in polynomial time.*

Proof. If $r = 0$, then $y_+ = y_-$, and $y_+ + y_- \leq 1$ forces $y_+ = y_- = 0$; then $x_i \leq 0$ for all $i \in V$. So the zero vector is the unique feasible point and $v = 0$. The native points form three slices. If $y_+ = y_- = 0$, then $x = 0$ and $(f, r) = (0, 0)$. If $(y_+, y_-) = (1, 0)$, the constraints on x say exactly that x is the indicator vector of a stable set S , and $(f, r) = (-|S|, 1)$. If $(y_+, y_-) = (0, 1)$, likewise $(f, r) = (-|S|, -1)$. Minimizing $f + \lambda r + \rho|r|$ over the three slices gives $\min\{0, \rho - \alpha(G) + \lambda, \rho - \alpha(G) - \lambda\}$, which is the formula for $L_\rho(\lambda)$. Its supremum over λ is attained at $\lambda = 0$, which gives D_ρ , and $D_\rho = 0$ exactly when $\rho \geq \alpha(G)$; the same holds for $L_\rho(0)$, so $\rho_* = \rho_*(0) = \alpha(G)$.

At $\rho = \alpha(G)$, $L_\rho(\lambda) = \min\{0, -|\lambda|\}$ equals 0 only for $\lambda = 0$, and a maximum stable set with $y_+ = 1$ then has penalized value 0. At multiplier zero, every infeasible point has value $\rho - |S|$ for a stable set S , which is positive for all such points exactly when $\rho > \alpha(G)$.

All coefficients and right-hand sides lie in $\{-1, 0, 1\}$, and computing $\alpha(G)$ is NP-hard, so computing ρ_* is strongly NP-hard. Deciding $\alpha(G) \leq c$ for an input integer c is the complement of the NP-complete stable-set decision problem, and a stable set of size $c + 1$ is a polynomial certificate that $\rho = c$ is not value-exact. For fixed c there are $O(|V|^{c+1})$ subsets of size $c + 1$ to check. $\square$

Unlike the binary box, where linear optimization over X is trivial, here linear optimization over the native set is already the maximum stable-set problem. We use this family only for strong NP-hardness of exact computation. It cannot exclude every polynomial factor as Theorem 6.4 does: since $1 \leq \alpha(G) \leq |V| \leq N$, the output $\hat{\rho} = |V|$ is always a sufficient calibration within factor $|V| \leq N$. The numeric family (38) is what yields arbitrary polynomial factors.

6.5 Relation to prior work

In their conclusion, Lefebvre and Schmidt [29] ask whether the smallest penalty parameter that closes the duality gap can be computed in polynomial time, and conjecture a negative answer. Their gap is defined through the supremum over multipliers (their Definition 1), which is value-exactness here, and their question concerns MILPs and MIQPs. The binary-box family (38) is a pure binary integer linear program, so it lies within the scope of their question. Theorem 6.4(a) confirms their conjecture unless $P = NP$, even for approximation within any polynomial factor, in growing binary dimension. Section 1.2 compares the results with earlier hardness results for choosing penalties.

7 Right-hand-side perturbations

The coefficient in Theorem 3.1 explodes because the right-hand side of the relaxed equality lies within δ_d of the boundary of the residual image of an improving slice, that is, of the set of right-hand sides attainable on that slice. This section shows that a margin from such boundaries controls the coefficient: if the perturbed problem is feasible and its right-hand side keeps distance more than γ from the boundaries of all residual images, the coefficient M/γ is exact at multiplier zero (Lemma 7.1). A right-hand side sampled from a fine rational grid has such a margin with high probability, which gives a coefficient of polynomial encoding length (Theorem 7.3). The principle is classical in conditioning theory [11, 15]: a right-hand side far from the boundaries of the convex residual images is far from any change in the set of feasible integer assignments, and random perturbations rarely come close to such boundaries. We prove the cube estimate and its transfer to the grid; the Gaussian variant in Section 7.7 uses Dunagan et al. [15, Lemma 2.3.4].

Perturbing the right-hand side changes the problem. All guarantees in this section concern the sampled instance, which may be infeasible and may have a different optimal integer assignment from the original one.

7.1 Setting

Let $\mathcal{Z}$ be a finite index set with $|\mathcal{Z}| \leq K$, where $K \geq 1$. For each $z \in \mathcal{Z}$, let X_z be a nonempty compact convex subset of a Euclidean space (the slice), let $r_z : X_z \rightarrow \mathbb{R}^m$ be affine, with $m \geq 1$, and let $f_z : X_z \rightarrow \mathbb{R}$ be continuous and convex. Bounds $\underline{f} \leq \bar{f}$ with

$$\underline{f} \leq f_z(x) \leq \bar{f} \quad (z \in \mathcal{Z}, x \in X_z)$$

are supplied, and $M = \bar{f} - \underline{f}$ denotes the objective range. The slices, maps and objectives are fixed before any sampling. For a right-hand side $b \in \mathbb{R}^m$, let

$$\begin{aligned} v(b) &= \min\{f_z(x) : z \in \mathcal{Z}, x \in X_z, r_z(x) = b\}, \\ p_\rho(b) &= \min_{z \in \mathcal{Z}, x \in X_z} \{f_z(x) + \rho \|r_z(x) - b\|_\infty\}, \end{aligned} \tag{47}$$

with $v(b) = +\infty$ if no point is feasible. We call b *feasible* if $v(b) < \infty$, and we write $v_z(b)$ for the minimum of f_z over $\{x \in X_z : r_z(x) = b\}$ (with $v_z(b) = +\infty$ if this set is empty).

This is the setting of Section 2: the native set is the disjoint union of the slices, the objective is f_z on slice z , the residual is $r_z(x) - b$, and the norm is the infinity norm. In particular, $p_\rho(b) = L_\rho(0)$ for this instance. For a feasible b , we write $\rho_*[b]$ for the least penalty of this instance, with the multiplier optimized; the square brackets distinguish it from the fixed-multiplier threshold $\rho_*(\lambda)$. For infeasible b the least penalty is not defined; where the distribution of $\rho_*[b]$ over random b is discussed, any nonnegative value may be assigned to infeasible b .

For example, in the model of Section 4 with convex slices, each integer assignment z with a nonempty slice gives such a slice, with f_z the objective and r_z the left-hand side of the linking equality for this fixed z . Then K can be taken as the product of the numbers of allowed values of the integer variables, so $\log K$ is polynomial in the input length, but K itself can be exponential in the number of integer variables. The present setting needs neither a Slater condition nor quadratic data.

The *residual image* $C_z = r_z(X_z)$ is the set of right-hand sides attainable on slice z ; it is nonempty, compact and convex, and the problem with right-hand side b has a feasible point in slice z exactly when $b \in C_z$. Boundaries ∂C are topological boundaries in $\mathbb{R}^m$. Thus a closed convex set with empty interior, such as a singleton, is its own boundary, and every right-hand side in it has margin zero: arbitrarily small perturbations leave it. Distances are $\text{dist}_\infty(b, S) = \inf_{y \in S} \|b - y\|_\infty$, and similarly dist_2 , with distance $+\infty$ to the empty set. The latter convention matters only in Lemma 7.2, where $C = \emptyset$ or $C = \mathbb{R}^m$ is allowed.

7.2 A deterministic boundary-margin bound

The idea is a convex repair. Suppose every residual image has boundary distance greater than γ from b . On a slice whose image contains b , the whole cube $b + [-\gamma, \gamma]^m$ then consists of attainable right-hand sides; in particular, this image is full-dimensional. A native point x at residual distance $t = \|r_z(x) - b\|_\infty > 0$ can be repaired by mixing it with a point x' whose residual lies on the far side of b at distance γ . The mixture has residual exactly b , and convexity bounds the cost of the repair by $(M/\gamma)t$. On a slice whose image does not contain b , every residual is at distance more than γ from b , so the penalty $(M/\gamma)t$ exceeds the whole objective range.

Lemma 7.1. *In the setting of Section 7.1, let $b \in \mathbb{R}^m$ be feasible and let $\gamma > 0$ satisfy $\text{dist}_\infty(b, \partial C_z) > \gamma$ for every $z \in \mathcal{Z}$. Then $p_{M/\gamma}(b) = v(b)$, that is, the coefficient M/γ is exact at multiplier zero for right-hand side b ; in particular $\rho_*[b] \leq M/\gamma$. Every coefficient $\rho > M/\gamma$ is minimizer-exact at multiplier zero.*

Proof. We show that $f_z(x) + (M/\gamma)\|r_z(x) - b\|_\infty \geq v(b)$ for every slice z and every $x \in X_z$, first on slices whose image contains b and then on the others.

Slices with $b \in C_z$. First, $b + [-\gamma, \gamma]^m \subseteq C_z$. Otherwise some y with $\|y - b\|_\infty \leq \gamma$ lies outside C_z . Since C_z is closed and contains b , the last point of C_z on the segment from b to y lies on ∂C_z , at infinity-norm distance at most γ from b , a contradiction.

Now let $x \in X_z$ with $t = \|r_z(x) - b\|_\infty > 0$. The point $b - (\gamma/t)(r_z(x) - b)$ lies in $b + [-\gamma, \gamma]^m \subseteq C_z$, so some $x' \in X_z$ satisfies

$$r_z(x') = b - \frac{\gamma}{t}(r_z(x) - b).$$

The convex combination $x_b = (\gamma x + t x')/(\gamma + t)$ lies in X_z , and since r_z is affine, $r_z(x_b) = b$. Convexity of f_z and $f_z(x') \leq \bar{f}$ give

$$v_z(b) \leq f_z(x_b) \leq \frac{\gamma}{\gamma + t} f_z(x) + \frac{t}{\gamma + t} \bar{f}.$$

Rearranging, and using $v(b) \leq v_z(b)$ and $\bar{f} - v_z(b) \leq \bar{f} - \underline{f} = M$,

$$f_z(x) \geq v_z(b) - \frac{\bar{f} - v_z(b)}{\gamma} t \geq v(b) - \frac{M}{\gamma} t.$$

This inequality also holds at $t = 0$, since then x is feasible and $f_z(x) \geq v(b)$.

Slices with $b \notin C_z$. A closest point of the compact set C_z to b lies on ∂C_z , so $\text{dist}_\infty(b, C_z) = \text{dist}_\infty(b, \partial C_z) > \gamma$. Hence every $x \in X_z$ has $t = \|r_z(x) - b\|_\infty > \gamma$ and

$$f_z(x) + \frac{M}{\gamma} t \geq \underline{f} + M = \bar{f} \geq v(b).$$

Thus $p_{M/\gamma}(b) \geq v(b)$, and a feasible optimum attains $v(b)$ with zero penalty, so equality holds. If $\rho > M/\gamma$, every point with nonzero residual has penalized value strictly greater than $v(b)$, so the minimizers are exactly the optimal feasible points. $\square$

The argument needs neither differentiability nor computed multipliers. It is a right-hand-side analogue of Lemma 2.6: the margin γ replaces the multiplier bound on slices whose image contains b by the convex repair, and it gives the residual separation γ on the other slices. The relation between exact penalties and multiplier bounds in convex optimization is classical; see Friedlander and Tseng [18, Theorem 4.2] for a polar-gauge characterization under a nonempty compact solution set.

Convexity of the objective is essential. Take one slice $X = [-1, 1]$ with $r(x) = x$, right-hand side $b = 0$ and $f(x) = -\sqrt{|x|}$. The image $[-1, 1]$ has boundary distance one from b , and the objective range is one. For $0 < t \leq 1$, the balanced mixture of $x = t$ and $x = -t$ with equal weights has average penalized value $-\sqrt{t} + \rho t$, independently of the multiplier. This is negative for small t , so by Proposition 2.2, $D_\rho < 0 = v$ for every $\rho \geq 0$: no finite penalty is value-exact.

7.3 Convex-boundary tubes in a cube

The next lemma bounds the probability that a uniform point of a cube falls near the boundary of a closed convex set. The bound is linear in the tube width and does not depend on the set.

Lemma 7.2. *Let $Q = b_0 + [-\sigma, \sigma]^m$ with $\sigma > 0$, and let B be uniform on Q . For every closed convex set $C \subseteq \mathbb{R}^m$ and every $\gamma \geq 0$,*

$$\Pr\{\text{dist}_\infty(B, \partial C) \leq \gamma\} \leq \min\{1, 2m\gamma/\sigma\}. \quad (48)$$

The same bound holds with dist_2 in place of dist_∞ .

Proof. The plan is to cover the tube by an outer shell and an inner shell of the set, bound the volume of each shell inside Q one coordinate direction at a time, and let the shell width decrease to γ .

If $C = \emptyset$ or $C = \mathbb{R}^m$, the boundary is empty and the event is empty. Otherwise ∂C is nonempty. Fix $w > 0$ and let $S_i = [-w, w]e_i$ for $i = 1, \dots, m$. For a closed convex set E , the addition $E + S_i$ and the erosion $E \ominus S_i = \{x : x + S_i \subseteq E\}$ are again closed and convex.

One-dimensional fact. Let $I \subseteq \mathbb{R}$ be a closed convex set (empty, a point, a closed interval, a closed ray or $\mathbb{R}$) and let W be an interval of length 2σ . Then

$$|((I + [-w, w]) \setminus I) \cap W| \leq 2w, \quad |(I \setminus (I \ominus [-w, w])) \cap W| \leq 2w,$$

where $|\cdot|$ is length. Indeed, addition extends I by w at each finite endpoint, and erosion removes at most w at each finite endpoint (possibly deleting a short interval entirely); there are at most two endpoints.

Shell volumes. On each line parallel to e_i , the section of $E + S_i$ is the section of E plus $[-w, w]$, the section of $E \ominus S_i$ is the section of E eroded by $[-w, w]$, and the section of Q is an interval of length 2σ . The one-dimensional fact and Fubini's theorem over the $(m-1)$ -dimensional face of Q orthogonal to e_i give

$$\begin{aligned}\text{vol}(((E + S_i) \setminus E) \cap Q) &\leq 2w(2\sigma)^{m-1}, \\ \text{vol}((E \setminus (E \ominus S_i)) \cap Q) &\leq 2w(2\sigma)^{m-1}.\end{aligned}$$

We measure set differences intersected with Q , which have finite volume. Successive additions of $S_1, \dots, S_m$ to C give $C + [-w, w]^m$. Successive erosions give $C \ominus [-w, w]^m$, because $(E \ominus S) \ominus T = E \ominus (S + T)$. The intermediate sets are closed and convex, so the bounds apply at each step, and telescoping over the m steps gives

$$\text{vol}(((C + [-w, w]^m) \setminus C) \cap Q) \leq 2mw(2\sigma)^{m-1}, \quad \text{vol}((C \setminus (C \ominus [-w, w]^m)) \cap Q) \leq 2mw(2\sigma)^{m-1}.$$

Tube containment. Let $w > \gamma$ and let $\text{dist}_\infty(x, \partial C) \leq \gamma$. Since ∂C is closed, there is $y \in \partial C$ with $\|x - y\|_\infty \leq \gamma$. As $y \in C$, we get $x \in C + [-w, w]^m$. As y is a boundary point, there is an exterior point $u \notin C$ with $\|u - y\|_\infty < w - \gamma$; then $\|u - x\|_\infty < w$, so $x + [-w, w]^m \not\subseteq C$ and $x \notin C \ominus [-w, w]^m$. Hence the tube lies in the union of the two shells. Dividing their total volume bound $4mw(2\sigma)^{m-1}$ by $\text{vol}(Q) = (2\sigma)^m$ gives probability at most $2mw/\sigma$. Letting $w \downarrow \gamma$ proves (48), including $\gamma = 0$. Only closedness and convexity of C were used, so unbounded and lower-dimensional sets are covered.

Finally, $\|\cdot\|_\infty \leq \|\cdot\|_2$, so the Euclidean event is contained in the infinity-norm event. $\square$

The coefficient $2m$ is sharp for bounds of the form $c_m \gamma / \sigma$. Let $C = b_0 + [-a, a]^m$ with $0 < a < \sigma$, and let $0 < \gamma \leq a$ with $a + \gamma \leq \sigma$. The tube is the cube of radius $a + \gamma$ minus the open cube of radius $a - \gamma$, both centred at b_0 and contained in Q , so its probability is $((a + \gamma)^m - (a - \gamma)^m) / \sigma^m$. Its derivative at $\gamma = 0$ is $(2m/\sigma)(a/\sigma)^{m-1}$, which tends to $2m/\sigma$ as $a \uparrow \sigma$.

7.4 Rational-grid exactness and its encoding

To obtain rational right-hand sides of controlled encoding length, we sample from a finite grid and couple the grid sample with a uniform point of Q . For an integer $q \geq 1$, let $h = 2\sigma/q$ and sample each coordinate of G independently and uniformly from the q cell centres

$$b_{0i} - \sigma + (j + 1/2)h, \quad j = 0, \dots, q - 1. \quad (49)$$

Let U be uniform on $[-h/2, h/2]^m$ and independent of G . Then $B = G + U$ is uniform on Q , and $\|B - G\|_\infty \leq h/2 = \sigma/q$. Since $\text{dist}_\infty(\cdot, \partial C)$ is one-Lipschitz in the infinity norm,

$$\{\text{dist}_\infty(G, \partial C) \leq \gamma\} \subseteq \{\text{dist}_\infty(B, \partial C) \leq \gamma + \sigma/q\},$$

and Lemma 7.2 gives, for every closed convex C and $\gamma \geq 0$,

$$\Pr\{\text{dist}_\infty(G, \partial C) \leq \gamma\} \leq \min\{1, 2m(\gamma/\sigma + 1/q)\}. \quad (50)$$

The rounding term $2m/q$ is necessary. Convex boundaries have Lebesgue measure zero, but grid points can lie on them with positive probability: for $m = 1$, an interval whose two endpoints are distinct grid points has boundary probability $2/q$ at $\gamma = 0$, exactly the correction in (50).

Theorem 7.3. Assume the setting of Section 7.1. Let $b_0 \in \mathbb{Q}^m$ and let $\sigma, \epsilon, \underline{f}, \bar{f}$ be rational with $\sigma > 0$ and $0 < \epsilon < 1$. Let G be sampled from the grid (49) with

$$q \geq \left\lceil \frac{4mK}{\epsilon} \right\rceil, \quad \gamma = \frac{\sigma\epsilon}{4mK}, \quad \bar{\rho} = \frac{M}{\gamma} = \frac{4mKM}{\sigma\epsilon}. \quad (51)$$

Then, with probability at least $1 - \epsilon$, either the sampled problem is infeasible or $p_{\bar{\rho}}(G) = v(G)$. On the same event, if the sampled problem is feasible, every coefficient $\rho > \bar{\rho}$ is minimizer-exact at multiplier zero.

Proof. By (50) and a union bound over the at most K images,

$$\Pr\{\text{dist}_{\infty}(G, \partial C_z) \leq \gamma \text{ for some } z \in \mathcal{Z}\} \leq 2mK \left(\frac{\gamma}{\sigma} + \frac{1}{q} \right) \leq 2mK \left(\frac{\epsilon}{4mK} + \frac{\epsilon}{4mK} \right) = \epsilon.$$

On the complementary event every boundary distance exceeds γ , and Lemma 7.1 applies whenever G is feasible. $\square$

In words: with probability at least $1 - \epsilon$, the grid sample has boundary distance greater than $\gamma = \sigma\epsilon/(4mK)$ from every residual image; then either no slice is feasible, or $\rho_*[G] \leq \bar{\rho} = M/\gamma$.

Remark 7.4 (Encoding). Choose q as the smallest power of two with $q \geq \lceil 4mK/\epsilon \rceil$. Then $\log_2 q = O(1 + \log m + \log K + \log(1/\epsilon))$, and exact uniform sampling uses $m \log_2 q$ independent random bits. The grid points and $\bar{\rho}$ have encoding length polynomial in the supplied rational data $b_0, \sigma, \epsilon, \underline{f}, \bar{f}$, in m and in $\log K$; an integer coefficient strictly larger than $\bar{\rho}$, such as $\lfloor \bar{\rho} \rfloor + 1$, has the same bound, although $\bar{\rho}$ is numerically of order K . Neither computing $\bar{\rho}$ nor sampling requires enumerating the slices. The construction does not check the structural assumptions of Section 7.1 and does not solve the sampled problem.

Remark 7.5 (Endpoint grid). Theorem 7.3 also holds for the grid with $q + 1$ points $b_{0i} - \sigma + jh$, $0 \leq j \leq q$, per coordinate. Its jitter cells tile the cube of radius $\sigma + h/2$, so Lemma 7.2 for that cube gives

$$\Pr\{\text{dist}_{\infty}(G, \partial C) \leq \gamma\} \leq \min \left\{ 1, \frac{2m(\gamma + h/2)}{\sigma + h/2} \right\} \leq \min\{1, 2m(\gamma/\sigma + 1/q)\}.$$

With $q + 1$ points per coordinate, the random-bit count of Remark 7.4 no longer applies. For either grid the Euclidean boundary event obeys the same estimate, but it pairs only with a Euclidean penalty: a Euclidean margin greater than γ does not give an infinity-norm margin greater than γ .

7.5 Feasibility, local stability and the original instance

The infeasibility alternative. Theorem 7.3 cannot guarantee feasibility. For example, if no residual image meets Q , every sample is infeasible. If some slice satisfies $Q \subseteq C_{z_0}$, every grid point is feasible, and the theorem gives exactness at multiplier zero with probability at least $1 - \epsilon$ unconditionally. More generally, if the grid sample is feasible with probability at least $p_0 > 0$, then the conditional failure probability given feasibility is at most ϵ/p_0 .

This bound concerns the grid actually used; refining the grid can lower its feasibility probability. For a recipe valid for every grid size, suppose that B uniform on Q is feasible with probability at least a known rational $p_0 > 0$. Under the jitter coupling, if $B \in C_z$ and $G \notin C_z$, then the segment from G to B crosses ∂C_z , so $\text{dist}_{\infty}(B, \partial C_z) \leq \sigma/q$. Lemma 7.2 and a union bound give

$$\Pr\{G \text{ feasible}\} \geq p_0 - \frac{2mK}{q}.$$

Running the construction with $\epsilon p_0/2$ in place of ϵ (so $q \geq \lceil 8mK/(\epsilon p_0) \rceil$ and $\bar{\rho} = 8mKM/(\sigma \epsilon p_0)$) gives $\Pr\{G \text{ feasible}\} \geq p_0(1 - \epsilon/4)$, while the failure probability is at most $\epsilon p_0/2$. Hence the conditional failure probability given feasibility is at most $\epsilon/(2 - \epsilon/2) < \epsilon$.

Local stability. Let b be feasible with $\text{dist}_\infty(b, \partial C_z) > \gamma$ for every z . By the first step of the proof of Lemma 7.1, every residual image C_z that contains b contains the whole cube $b + [-\gamma, \gamma]^m$, while every other image has distance greater than γ from b . Hence the set of slices (integer assignments) that are feasible for the right-hand side is the same at every point of the open cube of radius γ about b . We use the smaller cube $b + [-\gamma/2, \gamma/2]^m$ because, by the one-Lipschitz property of the distance, every point b' of it keeps margin greater than $\gamma/2$. Applying the repair inequality of Lemma 7.1 at b' with margin $\gamma/2$ to an optimal point for b'' , and then with the roles exchanged, gives

$$|v_z(b') - v_z(b'')| \leq \frac{2M}{\gamma} \|b' - b''\|_\infty$$

for every slice z whose image contains b and all b', b'' in the cube. A minimum over the same finite collection of slice values keeps this Lipschitz bound, so v is $2M/\gamma$ -Lipschitz in the infinity norm on the cube. By Lemma 7.1 with margin $\gamma/2$, the coefficient $2M/\gamma$ is exact at multiplier zero at every point of the cube, and every larger coefficient is minimizer-exact at multiplier zero there. Lipschitz value bounds across discrete choices under uniform Slater conditions also appear in mixed-integer optimal control [23, Theorem 3].

The original instance. The local conclusion does not recover the optimum at b_0 . The chain of Section 3 fits the present setting with $m = 1$, two slices and $M = 1$: the slice with binary variable zero has residual image $[-1, 1]$ and cost 0, and the improving slice (binary variable one) has residual image $[\delta_d, 1]$ and cost -1 . The right-hand side 0 lies within δ_d of the boundary of $[\delta_d, 1]$, so Lemma 7.1 gives only coefficients above $1/\delta_d$, the zero-multiplier threshold of Section 3; this is why the coefficient explodes. Moving the right-hand side to δ_d makes the improving assignment feasible, and the optimum becomes -1 instead of 0. Short penalties after perturbation can therefore come with a different optimal integer assignment.

7.6 Numerical size: unavoidable tails and slice dependence

The coefficient $\bar{\rho} = 4mKM/(\sigma \epsilon)$ of Theorem 7.3 has short encoding but can be numerically large. The following examples show which parts of this size are unavoidable:

- (a) the order K/ϵ is necessary: some instances satisfy $\Pr\{\rho_*[B] > T\} \geq (K - 1)/T$ for $T \geq K$ (Example 7.6);
- (b) a lower bound of order K can hold almost surely, not just with small probability (Example 7.7);
- (c) two slices can give $\mathbb{E}\rho_*[B] = +\infty$ while $\mathbb{E}\log(1 + \rho_*[B])$ is finite (Example 7.8);
- (d) an exceptional grid point can have infinite threshold, so the high-probability guarantee cannot be turned into a bound on expectations (Example 7.9).

In all examples $m = 1$, so the norm is the absolute value, and $\underline{f} = -1$, $\bar{f} = 0$, $M = 1$. All thresholds are optimized thresholds $\rho_*[b]$ unless we say otherwise; since exactness at multiplier zero implies value-exactness, lower bounds on $\rho_*[b]$ also bound the zero-multiplier threshold. The examples list their slices explicitly, as the slice model allows; they are not derived from a succinct mixed-integer encoding.

The lower bounds use one balanced mixture (Proposition 2.2). Suppose a slice with cost -1 has a singleton residual image at distance $t > 0$ from b , and a cost-zero slice contains a point whose residual $r_z(x) - b$ has the opposite sign and magnitude $s > 0$. Put weight $s/(s+t)$ on the singleton point and weight $t/(s+t)$ on the cost-zero point. The residuals cancel, and the average penalized value is

$$\frac{s}{s+t}(-1 + 2\rho t) \quad (52)$$

for every multiplier. If $v(b) = 0$, this mixture shows $\rho_*[b] \geq 1/(2t)$.

Example 7.6 (The order K/ϵ is necessary). Let an anchor slice have residual image $[0, 1]$ and cost zero, and let $K - 1$ singleton slices, $K \geq 2$, have images $\{j/K\}$, $1 \leq j \leq K - 1$, and cost -1 . At every right-hand side $b \in (0, 1)$ that is not one of the points j/K , the primal value is zero, and (52) with the nearest singleton at distance t gives $\rho_*[b] \geq 1/(2t)$. Let B be uniform on $[0, 1]$. For $T \geq K$, the intervals $|B - j/K| < 1/(2T)$ are disjoint, lie in $(0, 1)$ and have total length $(K - 1)/T$. Consequently

$$\Pr\{\rho_*[B] > T\} \geq \frac{K - 1}{T} \quad (T \geq K). \quad (53)$$

Let $0 < \epsilon < (K - 1)/K$. For $K \leq T < (K - 1)/\epsilon$, (53) gives $\Pr\{\rho_*[B] > T\} > \epsilon$, and for $T < K$ the same holds by monotonicity. Hence the $(1 - \epsilon)$ -quantile of $\rho_*[B]$ is at least $(K - 1)/\epsilon$. Here $b_0 = 1/2$, $\sigma = 1/2$ and $M = 1$, so Theorem 7.3 gives the coefficient $8K/\epsilon$; the order K/ϵ cannot be improved, even though every sample is feasible and the multiplier is optimized.

The grid sample of Theorem 7.3 behaves the same way. On $[0, 1]$ the cell centres are $(i + 1/2)/q$; let K divide q , so that no centre is a singleton image. If $q \geq 2K$, each singleton image j/K has two adjacent centres at distance $1/(2q)$, and these $2(K - 1)$ centres are distinct; hence $\Pr\{\rho_*[G] \geq q\} \geq 2(K - 1)/q$. More generally, for $K < T \leq q$, each singleton image has at least $q/T - 1$ centres at distance less than $1/(2T)$, these sets are disjoint for different j , and at each such centre $\rho_*[G] > T$. Hence $\Pr\{\rho_*[G] > T\} \geq (K - 1)(1/T - 1/q)$. If $\epsilon \leq (K - 1)/(2K)$ and $q \geq 4K/\epsilon$, then $T = (K - 1)/(2\epsilon)$ satisfies $K \leq T \leq q$, and $\Pr\{\rho_*[G] > T\} \geq 2\epsilon - (K - 1)/q > \epsilon$. So the $(1 - \epsilon)$ -quantile of $\rho_*[G]$ is at least $(K - 1)/(2\epsilon)$, and the order K/ϵ is necessary for the grid result as well.

Example 7.7 (An almost-sure slice-count obstruction). Keep the anchor slice with image $[0, 1]$ and cost zero, and use $J + 1$ singleton slices, $J \geq 1$, with images $\{j/J\}$, $j = 0, \dots, J$, and cost -1 . There are $K = J + 2$ slices. Let $b \in (0, 1)$ lie strictly between the neighbouring singleton images j/J and $(j + 1)/J$, and let $g_1 = b - j/J > 0$ and $g_2 = (j + 1)/J - b > 0$ be its distances to them, so $g_1 + g_2 = 1/J$. The primal value is zero. Put weight $g_2/(g_1 + g_2)$ on the lower singleton and $g_1/(g_1 + g_2)$ on the upper one. The residuals cancel, the cost is -1 , and the mean absolute residual is

$$\frac{2g_1g_2}{g_1 + g_2} = 2Jg_1g_2 \leq \frac{1}{2J}.$$

Hence $\rho_*[b] \geq 2J = 2(K - 2)$ for every $b \in (0, 1)$ other than the singleton images, that is, for almost every $b \in [0, 1]$.

Example 7.8 (Infinite mean with two slices). Take an anchor slice with image $[-1, 1]$ and cost zero and a singleton slice with image $\{0\}$ and cost -1 . For $b \in (-1, 1) \setminus \{0\}$, the primal value is zero, and (52) with $t = |b|$ gives $\rho_*[b] \geq 1/(2|b|)$. Conversely, at $\rho = 1/(2|b|)$ and multiplier $\lambda = -\rho \operatorname{sign}(b)$, the singleton's penalized value is $-1 + \lambda(-b) + \rho|b| = -1 + 2\rho|b| = 0$, and the anchor's penalized value $\lambda e + \rho|e|$ at residual e is nonnegative. Hence

$$\rho_*[b] = \frac{1}{2|b|}. \quad (54)$$

The zero-multiplier threshold is instead $1/|b|$. For B uniform on $[-1, 1]$, we get $\mathbb{E}\rho_*[B] = +\infty$, while $\mathbb{E}\log(1 + \rho_*[B]) < \infty$ because $\log(1 + 1/(2|b|))$ is integrable near zero.

Example 7.9 (An exceptional grid point with infinite threshold). Take a convex quadratic slice $\{(\alpha, \beta) : 0 \leq \alpha \leq 1, \alpha^2 \leq \beta \leq 1\}$ with cost $-\alpha$ and residual β , so its image is $[0, 1]$, and a singleton slice with image $\{-1\}$ and cost zero. At right-hand side 0 the constraint $\beta = 0$ forces $\alpha = 0$, so the primal value is zero. For $0 < u \leq 1$, mix the point $(\alpha, \beta) = (u, u^2)$ with weight $1/(1 + u^2)$ and the singleton with weight $u^2/(1 + u^2)$. The residuals cancel, the cost is $-u/(1 + u^2)$, and the mean absolute residual is $2u^2/(1 + u^2)$. By Corollary 2.3, $\rho_*[0]$ is at least the ratio $1/(2u)$ for every such u , so $\rho_*[0] = +\infty$: no finite penalty is value-exact. A rational grid containing 0, for instance the grid (49) with $b_0 = 0$ and odd q , gives this right-hand side positive probability. Hence $\mathbb{E}\rho_*[G] = +\infty$ (with any nonnegative value assigned to infeasible samples), and no almost-sure bound on the threshold or its encoding holds. The margin condition of Lemma 7.1 excludes this point, since 0 lies on the boundary of $[0, 1]$. The refined Slater condition also fails there: on $\beta = 0$ the inequality $\alpha^2 \leq \beta$ cannot hold strictly.

7.7 A Gaussian companion in the Euclidean norm

Lemma 7.1 and its proof hold for any norm on $\mathbb{R}^m$, provided the same norm defines both the margin and the penalty: replace the cube $b + [-\gamma, \gamma]^m$ by the ball of radius γ . For Gaussian noise we use the Euclidean norm for both. There is no grid here, so this result has no encoding statement.

Proposition 7.10. *Assume the setting of Section 7.1, but with the Euclidean norm in the penalty of (47). Let $B = b_0 + \sigma\xi$ with $\xi \sim \mathcal{N}(0, I_m)$, $\sigma > 0$ and $0 < \epsilon < 1$. Then for every $\gamma \geq 0$,*

$$\Pr\{\min_{z \in \mathcal{Z}} \text{dist}_2(B, \partial C_z) \leq \gamma\} \leq \frac{8Km^{1/4}\gamma}{\sigma}. \quad (55)$$

Consequently, with probability at least $1 - \epsilon$, either the sampled problem is infeasible or the coefficient

$$\frac{8MKm^{1/4}}{\sigma\epsilon} \quad (56)$$

is exact at multiplier zero for the Euclidean penalty; every larger coefficient is then minimizer-exact at multiplier zero.

Proof. First let C be a nonempty compact convex set with nonempty interior. Dunagan et al. [15, Lemma 2.3.4] bound the probabilities of the outer tube $\{\text{dist}_2(B, \partial C) \leq \gamma\} \setminus C$ and of the inner tube $\{\text{dist}_2(B, \partial C) \leq \gamma\} \cap C$ each by $4m^{1/4}\gamma/\sigma$. Hence $\Pr\{\text{dist}_2(B, \partial C) \leq \gamma\} \leq 8m^{1/4}\gamma/\sigma$.

Now let C be nonempty, compact and convex with empty interior, so $\partial C = C$. For $\eta > 0$, the parallel body $C_\eta = C + \eta\mathbb{B}_2$ is compact and convex with nonempty interior. We claim $\text{dist}_2(x, \partial C_\eta) \leq |\text{dist}_2(x, C) - \eta|$. Let p be the projection of x onto C , and let u be a unit vector in the normal cone of C at p : take $u = (x - p)/\|x - p\|_2$ if $x \notin C$, and a unit normal to the affine hull of C if $x \in C$ (it exists because C has empty interior). Then $p + su$ has projection p and distance s from C for every $s \geq 0$. So $p + \eta u \in C_\eta$, while $p + su \notin C_\eta$ for $s > \eta$; thus $p + \eta u \in \partial C_\eta$. Since $x = p + \text{dist}_2(x, C)u$, the claim follows. Hence $\text{dist}_2(x, C) \leq \gamma$ implies $\text{dist}_2(x, \partial C_\eta) \leq \gamma + \eta$, and the first case gives

$$\Pr\{\text{dist}_2(B, \partial C) \leq \gamma\} \leq \Pr\{\text{dist}_2(B, \partial C_\eta) \leq \gamma + \eta\} \leq \frac{8m^{1/4}(\gamma + \eta)}{\sigma}.$$

Letting $\eta \downarrow 0$ gives the same bound as for convex bodies.

A union bound over the at most K images proves (55). With $\gamma = \sigma\epsilon/(8Km^{1/4})$, the right-hand side equals ϵ . On the complementary event, every Euclidean boundary distance exceeds γ , and the Euclidean version of Lemma 7.1 shows that M/γ , which equals (56), is exact at multiplier zero whenever B is feasible, and that every larger coefficient is minimizer-exact at multiplier zero. $\square$

8 Discussion and open questions

The results separate three issues about exact norm penalties: whether a sufficient coefficient has a short encoding, whether a nearly least one can be found efficiently, and how the position of the right-hand side affects both. This section draws lessons for choosing penalties, states the scope of the results and lists open questions.

Lessons for choosing penalties.

- *Large penalties come from nearly feasible improving assignments.* If an integer assignment that would improve the objective has a point that comes very close to satisfying the relaxed equalities, then by (2) the zero-multiplier threshold is at least the objective gain divided by the residual norm of that point. The least penalty is at least half this ratio when, in addition, some native point of cost at most v has a residual pointing in the opposite direction, because a balanced mixture of the two points cancels every multiplier; without such points the least penalty can be small (for one relaxed equality with one-sided residuals it is zero). Slater regularity on the equality-feasible slices prevents neither effect: in Theorem 3.1 every integer slice has points with slack $1/8$, yet the improving slice comes within $\delta_d = 2^{-2^d}$ of the relaxed equality.
- *A safe coefficient follows from two separate bounds.* By Lemma 2.6, a bound Λ on multipliers on the equality-feasible slices and a lower bound δ on the residual of the equality-infeasible slices, together with objective bounds, give a coefficient that is exact at multiplier zero. Objective bounds are often available, and Slater points control Λ (Sections 4 and 5). The separation δ is the quantity that can be tiny.
- *Optimizing the multiplier does not remove the growth.* In Theorem 3.1 the least penalty is half the zero-multiplier threshold, and both are doubly exponential in d . With one relaxed equality, optimizing the multiplier reduces the required coefficient by at most a factor of two when both one-sided rates are nonnegative (Proposition 2.4). For several relaxed equalities we prove no such factor bound.
- *Safe is easy, nearly least is hard.* On the instances of Theorem 6.4, the coefficient $\rho = 1$ is exact at multiplier zero and the optimal solution is known, yet no polynomial-time algorithm finds a sufficient coefficient within a polynomial factor of the least penalty unless $P = NP$. The difficulty lies in testing whether a given coefficient is exact (Remark 6.9), and rescaling the relaxed equality does not help (Remark 6.7). The hardness is worst-case and weak, and it uses growing binary dimension.
- *Random right-hand sides avoid the bad case but change the problem.* A random right-hand side avoids the boundaries that cause large penalties with high probability (Theorem 7.3), but the guarantee is for the perturbed problem, whose optimal integer assignment can differ from the original one (Section 7.5). The coefficient grows like K/ϵ , this order is necessary, and the least penalty has heavy tails (Section 7.6).

Scope. All results except one lower-bound extension concern norm penalties with a fixed norm and a fixed residual map, and coefficients in ordinary binary encoding. The extension is that the lower bound of Theorem 3.1 holds for every penalty function with at most linear growth near zero, including the squared (proximal) augmentation (Section 3.4). Apart from that lower bound, nothing is claimed about the classical augmented Lagrangian with a squared penalty, about solving the penalized subproblems or about solver performance. Section 3.4 also explains why rescaling the relaxed equality, an instance-dependent fractional-power augmentation or succinct exponent notation changes the question. The upper bounds of Sections 4 and 5 are existence bounds whose constants are not evaluated, and the coefficients they give may be numerically huge.

Open questions

1. *The fixed-count gap.* Within the class of Assumption 4.1 with at most k native inequalities that are nonlinear in the continuous variables, the chain variant of Remark 3.5 needs $\Omega(N2^k)$ bits, while Theorem 5.1 gives $N^{O(k+1)}$ bits. Is the worst case $N2^{\Theta(k)}$, $N^{\Theta(k)}$ or in between?
2. *Without the refined Slater condition.* Suppose a boxed instance with convex quadratic slices has a finite least penalty, but the refined Slater condition fails on some equality-feasible slice. Does a sufficient coefficient with $(N+1)2^{O(n)}$ or $N^{O(k+1)}$ bits still exist? If not, what is the worst-case encoding length of ρ_* ?
3. *Nonconvex quadratic native constraints.* Under assumptions that guarantee finite exactness for nonconvex slices, such as those of Lefebvre and Schmidt [29, Theorem 17], do encoding bounds of the form $(N+1)2^{O(n)}$ still hold for degree-two data, or can nonconvexity force longer coefficients at fixed n ?
4. *Calibration in fixed dimension.* Our hardness results use a growing number of binary variables. For a fixed number of variables, or for a fixed number of integer variables with linear native constraints, can ρ_* or $\rho_*(0)$ be computed, or approximated within a constant factor, in polynomial time?
5. *Sharper perturbation bounds.* Can K in Theorem 7.3 be replaced by the number of slices whose residual images come near the right-hand side? For which noise radii does the perturbed problem keep the original optimal integer assignment with high probability while still admitting an explicit short coefficient?
6. *Usable conservative coefficients.* Can the constants in Theorems 4.2 and 5.1 be evaluated, or replaced by instance-dependent certificates, for example computable lower bounds on the separations of equality-infeasible slices and on Slater margins, so that the conservative coefficients become usable in computation?

Supplementary checks. Supplementary scripts (folder `verification`) check, with exact rational or symbolic arithmetic, the finite envelopes and thresholds of the lower-bound instances (including the seed variant of Remark 3.5), the determinant identities of Section 5 and a degenerate example (no Slater point, singular objective Hessian, divergent regularized multipliers) for the regularization in the proof of Lemma 5.2, the calibration formulas and thresholds, and selected grid, tail and feasibility-transfer identities from Section 7. These checks support the algebra; the proofs establish the general claims.

References

- [1] Shabbir Ahmed, Filipe Goulart Cabral, and Bernardo Freitas Paulo da Costa. Stochastic Lipschitz dynamic programming. *Mathematical Programming*, 191(2):755–793, 2022. doi: 10.1007/s10107-020-01569-z. Cited version: arXiv:1905.02290v2, May 23, 2019.
- [2] Edoardo Alessandrini, Sergi Ramos-Calderer, Ingo Roth, Emiliano Traversi, and Leandro Aolita. Alleviating the quantum Big-M problem. *npj Quantum Information*, 11:125, 2025. doi: 10.1038/s41534-025-01067-0. URL <https://arxiv.org/abs/2307.10379v4>. Cited version: arXiv:2307.10379v4, July 30, 2025.
- [3] Edoardo Alessandrini, Sergi Ramos-Calderer, Michel Krispin, Fritz Schinkel, Stefan Walter, Martin Kliesch, Leandro Aolita, and Ingo Roth. Scalable determination of penalization weights for constrained optimizations on approximate solvers, 2026. URL <https://arxiv.org/abs/2604.02416v1>. arXiv:2604.02416v1.
- [4] Saugata Basu. Algorithms in real algebraic geometry: A survey, 2014. URL https://www.math.purdue.edu/~sbasu/raag_survey2011_final-sep4-2014.pdf. Cited version: author version dated September 4, 2014 (arXiv:1409.1534). Published in J.-P. Monnier, K. Bekka, G. Fichou and R. Quarez (eds.), *Real Algebraic Geometry, Panoramas et Synthèses 51*, Société Mathématique de France, 2017, pp. 107–153.
- [5] Saugata Basu and Marie-Françoise Roy. Bounding the radii of balls meeting every connected component of semi-algebraic sets. *Journal of Symbolic Computation*, 45(12):1270–1279, 2010. doi: 10.1016/j.jsc.2010.06.009. URL https://www.math.purdue.edu/~sbasu/jsc_final-06-05-10.pdf. Cited version: final author manuscript, June 5, 2010.
- [6] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*, volume 10 of *Algorithms and Computation in Mathematics*. Springer, 2 edition, 2006. doi: 10.1007/3-540-33099-2. URL <https://www.math.purdue.edu/~sbasu/bpr-posted1.pdf>.
- [7] Yasmine Beck, Daniel Bienstock, Martin Schmidt, and Johannes Thürauf. On a computationally ill-behaved bilevel problem with a continuous and nonconvex lower level. *Journal of Optimization Theory and Applications*, 198(1):428–447, 2023. doi: 10.1007/s10957-023-02238-9. URL <https://optimization-online.org/wp-content/uploads/2022/02/nearly-feasible-bilevel-preprint.pdf>. Cited version: author manuscript dated May 3, 2023.
- [8] Avinash Bhardwaj, Vishnu Narayanan, and Abhishek Pathapati. Exact augmented Lagrangian duality for mixed integer convex optimization. *SIAM Journal on Optimization*, 34(2):1622–1645, 2024. doi: 10.1137/22M1526204. URL <https://arxiv.org/abs/2209.13326v2>. Cited version: arXiv:2209.13326v2, February 7, 2024.
- [9] Daniel Bienstock, Alberto Del Pia, and Robert Hildebrand. Complexity, exactness, and rationality in polynomial optimization. *Mathematical Programming*, 197(2):661–692, 2023. doi: 10.1007/s10107-022-01818-3. URL <https://arxiv.org/abs/2011.08347v5>. Cited version: arXiv:2011.08347v5, April 14, 2022.
- [10] Natasha L. Boland and Andrew C. Eberhard. On the augmented Lagrangian dual for integer programming. *Mathematical Programming*, 150(2):491–509, 2015. doi: 10.1007/s10107-014-0763-3. URL <https://optimization-online.org/wp-content/uploads/2013/01/3738.pdf>. Cited version: author manuscript posted on Optimization Online, January 2013.

- [11] Peter Bürgisser and Dennis Amelunxen. Robust smoothed analysis of a condition number for linear programming. *Mathematical Programming*, 131(1–2):221–251, 2012. doi: 10.1007/s10107-010-0346-x. URL <https://arxiv.org/abs/0803.0925v3>. Cited version: arXiv:0803.0925v3, January 22, 2010.
- [12] James V. Burke. An exact penalization viewpoint of constrained optimization. *SIAM Journal on Control and Optimization*, 29(4):968–998, 1991. doi: 10.1137/0329054.
- [13] M. Cordova, W. de Oliveira, and C. Sagastizábal. Revisiting augmented Lagrangian duals. *Mathematical Programming*, 196(1–2):235–277, 2022. doi: 10.1007/s10107-021-01703-5. Cited version: Optimization Online preprint, March 2020.
- [14] M. V. Dolgopoliuk. Augmented Lagrangian functions for cone constrained optimization: The existence of global saddle points and exact penalty property. *Journal of Global Optimization*, 71(2):237–296, 2018. doi: 10.1007/s10898-017-0603-0. Cited version: arXiv:1707.05747v6.
- [15] John Dunagan, Daniel A. Spielman, and Shang-Hua Teng. Smoothed analysis of condition numbers and complexity implications for linear programming. *Mathematical Programming*, 126(2):315–350, 2011. doi: 10.1007/s10107-009-0278-5. URL <https://www.cs.yale.edu/homes/spielman/Research/lpcond.pdf>. Cited version: author manuscript dated March 30, 2009.
- [16] I. I. Eremin. The penalty method in convex programming. *Soviet Mathematics Doklady*, 8: 459–462, 1966.
- [17] Mohammad Javad Feizollahi, Shabbir Ahmed, and Andy Sun. Exact augmented Lagrangian duality for mixed integer linear programming. *Mathematical Programming*, 161(1–2):365–387, 2017. doi: 10.1007/s10107-016-1012-8. URL https://www2.isye.gatech.edu/~xsun84/publications/EALD_optimizationonline.pdf. Cited version: author manuscript submitted August 13, 2015.
- [18] Michael P. Friedlander and Paul Tseng. Exact regularization of convex programs. *SIAM Journal on Optimization*, 18(4):1326–1350, 2007. doi: 10.1137/060675320. URL <https://optimization-online.org/wp-content/uploads/2006/11/1531.pdf>. Cited version: author manuscript dated November 18, 2006.
- [19] Christian Füllner and Steffen Rebennack. Stochastic dual dynamic programming and its variants: A review. *SIAM Review*, 67(3):415–539, 2025. doi: 10.1137/23M1575093.
- [20] Björn Geißler, Antonio Morsi, Lars Schewe, and Martin Schmidt. Penalty alternating direction methods for mixed-integer optimization: A new view on feasibility pumps. *SIAM Journal on Optimization*, 27(3):1611–1636, 2017. doi: 10.1137/16M1069687. Cited version: arXiv:1707.09891v1.
- [21] Dima Grigoriev and Dmitrii V. Pasechnik. Polynomial-time computing over quadratic maps I: Sampling in real algebraic sets. *Computational Complexity*, 14(1):20–52, 2005. doi: 10.1007/s00037-005-0189-7. URL <https://arxiv.org/abs/cs/0403008v3>. Cited version: arXiv:cs/0403008v3.
- [22] Xiaoyi Gu, Shabbir Ahmed, and Santanu S. Dey. Exact augmented Lagrangian duality for mixed integer quadratic programming. *SIAM Journal on Optimization*, 30(1):781–797, 2020. doi: 10.1137/19M1271695. URL <https://arxiv.org/abs/1907.00920v1>. Cited version: arXiv:1907.00920v1, July 1, 2019.

- [23] Martin Gugat and Falk M. Hante. Lipschitz continuity of the value function in mixed-integer optimal control problems. *Mathematics of Control, Signals, and Systems*, 29(1):Art. 3, 2017. doi: 10.1007/s00498-016-0183-4. URL <https://arxiv.org/abs/1612.04639v2>. Cited version: arXiv:1612.04639v2, January 10, 2017.
- [24] Nicolò Gusmeroli and Angelika Wiegele. EXPEDIS: An exact penalty method over discrete sets. *Discrete Optimization*, 44:100622, 2022. doi: 10.1016/j.disopt.2021.100622. URL <https://arxiv.org/abs/1912.09739v4>. Cited version: arXiv:1912.09739v4, January 3, 2021.
- [25] X. X. Huang and X. Q. Yang. A unified augmented Lagrangian approach to duality and exact penalization. *Mathematics of Operations Research*, 28(3):533–552, 2003. doi: 10.1287/moor.28.3.533.16395.
- [26] Jonas Kamminga and Dorian Rudolph. The pure-state consistency of local density matrices problem: In PSPACE and complete for a class between QMA and QMA(2). In *17th Innovations in Theoretical Computer Science Conference (ITCS 2026)*, volume 362 of *Leibniz International Proceedings in Informatics*, pages 83:1–83:23, 2026. doi: 10.4230/LIPIcs.ITCS.2026.83. URL <https://arxiv.org/abs/2411.03096v2>. Section numbers refer to the full version, arXiv:2411.03096v2, April 8, 2025.
- [27] Ed Klotz and Alexandra M. Newman. Practical guidelines for solving difficult mixed integer linear programs. *Surveys in Operations Research and Management Science*, 18(1–2):18–32, 2013. doi: 10.1016/j.sorms.2012.12.001.
- [28] Aljaž Krpan, Janez Povh, and Dunja Pucher. Quantum computing and the stable set problem. *Optimization Methods and Software*, 40(4):837–870, 2025. doi: 10.1080/10556788.2025.2490639. Cited version: arXiv:2405.12845v4.
- [29] Henri Lefebvre and Martin Schmidt. Exact augmented Lagrangian duality for nonconvex mixed-integer nonlinear optimization, 2025. URL <https://optimization-online.org/wp-content/uploads/2024/07/exact-penalty-for-minlp-1.pdf>. Manuscript dated December 15, 2025.
- [30] Puya Mirkarimi, David C. Hoyle, Ross Williams, and Nicholas Chancellor. Experimental demonstration of improved quantum optimization with linear Ising penalties. *New Journal of Physics*, 26(10):103005, 2024. doi: 10.1088/1367-2630/ad7e4a. URL <https://arxiv.org/abs/2404.05476v2>. Cited version: arXiv:2404.05476v2, December 16, 2024.
- [31] R. Tyrrell Rockafellar. *Convex Analysis*, volume 28 of *Princeton Mathematical Series*. Princeton University Press, Princeton, NJ, 1970.
- [32] R. Tyrrell Rockafellar. Augmented Lagrange multiplier functions and duality in nonconvex programming. *SIAM Journal on Control*, 12(2):268–285, 1974. doi: 10.1137/0312021.
- [33] R. Tyrrell Rockafellar and Roger J.-B. Wets. *Variational Analysis*, volume 317 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin, 1998. doi: 10.1007/978-3-642-02431-3.
- [34] Kaizhao Sun, Mou Sun, and Wotao Yin. Decomposition methods for global solution of mixed-integer linear programs. *SIAM Journal on Optimization*, 34(2):1206–1235, 2024. doi: 10.1137/22M1487321. Cited version: arXiv:2102.11980v4, January 1, 2024.

- [35] Zhen Wang, Mohammad Reza Hesamzadeh, Shudian Zhao, and Jan Kronqvist. A framework for eliminating paradoxical orders in European day-ahead electricity markets through mixed-integer linear programming strong duality, 2026. URL <https://arxiv.org/abs/2603.25490v1>. arXiv:2603.25490v1.
- [36] Willard I. Zangwill. Non-linear programming via penalty functions. *Management Science*, 13(5):344–358, 1967. doi: 10.1287/mnsc.13.5.344.
- [37] Shixuan Zhang and Xu Andy Sun. Stochastic dual dynamic programming for multistage stochastic mixed-integer nonlinear optimization. *Mathematical Programming*, 196(1–2):935–985, 2022. doi: 10.1007/s10107-022-01875-8. Cited version: arXiv:1912.13278v5.

A Remark on Example 13 of Lefebvre and Schmidt

Under the supremum-based Definition 1 of Lefebvre and Schmidt [29], their Example 13 is value-exact at every $\rho \geq 0$, although no finite multiplier attains the dual value. The example therefore shows nonattainment, not failure of value-exactness. It is a mixed-integer counterpart of Example 2.5.

Definition 1 of Lefebvre and Schmidt [29] says that a problem admits an exact penalty representation if the duality gap $v - D_\rho$ vanishes for some finite ρ , where D_ρ is the supremum over multipliers as in (1). In our terms, this means $\rho_* < \infty$. The source states that Example 13 admits no finite exact penalty representation. After renaming the variables (x_1, x_2, x_3) as (u, y, z) , the example has native set, objective and relaxed equality

$$X = \{(u, y, z) \in [-1, 1]^2 \times \{0, 1\} : u + 2z \leq 1, u^2 \leq y\}, \quad f = u - z/2, \quad r = y,$$

with $m = 1$ and the absolute value as norm. The equality $y = 0$ and the constraint $u^2 \leq y$ force $u = 0$, and then $2z \leq 1$ forces $z = 0$. So $(0, 0, 0)$ is the only feasible point and $v = 0$.

Value-exactness. Every native point has residual $y \geq u^2 \geq 0$. As for one-sided residuals in Section 2, every balanced mixture is then supported on F , and Proposition 2.2 gives $D_\rho = 0 = v$ for every $\rho \geq 0$. Thus $\rho_* = 0$, and the example satisfies Definition 1.

No attaining multiplier. Fix λ and ρ , and put $t = \lambda + \rho$. Since $y \geq 0$ on X , the penalized objective is $u - z/2 + ty$. For $-1 < u < 0$, the point $(u, u^2, 0)$ is native, and its penalized value $u + tu^2 = u(1 + tu)$ is negative whenever also $tu > -1$. Such u exist for every finite t . Hence $L_\rho(\lambda) < 0 = v$ for every finite λ and ρ : no multiplier attains D_ρ , and $\rho_*(\lambda) = +\infty$ for every λ . Multipliers whose penalized values approach v must therefore grow without bound; a bounded sequence of them would have a limit point attaining v , by continuity of L_ρ .

As a check, for $t \geq 1/2$ the value can be computed in closed form. The slice $z = 1$ contains only the point $(-1, 1, 1)$, with penalized value $t - 3/2$. On the slice $z = 0$, minimizing first over y gives $y = u^2$, and then $u = -1/(2t)$, with value $-1/(4t)$. Hence

$$L_\rho(\lambda) = \min\{t - 3/2, -1/(4t)\} \quad (t = \lambda + \rho \geq 1/2),$$

which tends to $0 = D_\rho$ as $\lambda \rightarrow +\infty$ and is negative for every finite λ .

Relation to the source's argument. The source reduces Example 13 to the multiplier $\lambda = 0$ by its Lemmas 2 and 3, and then shows that $L_\rho(0) < 0$ for every ρ . Its Lemma 3 (and its symmetric converse) transfers exactness attained at one fixed multiplier to another; it does not apply when the supremum is not attained. The source's Theorems 10 and 12 and Corollary 5, however, equate an exact penalty representation with attainment at a fixed multiplier, and its Theorem 14 concludes this attained notion. Under that notion the conclusion of Example 13 is correct, and the example still shows that Assumption 5 of Lefebvre and Schmidt [29] cannot be dropped from their Theorem 14: the equality-feasible slice $z = 0$ has no point with $y = 0$ and $u^2 < y$, and $L_\rho(\lambda) < v$ for all finite λ and ρ . The point concerns only Definition 1. It does not affect their Theorem 14, nor Theorem 3.1, whose dual supremum is attained.

For a genuine failure of value-exactness, see Example 7.9. Like Example 13, it violates the refined Slater condition. Unlike Example 13, its residuals take both signs, so balanced mixtures need not lie on F , and at right-hand side zero no finite penalty is value-exact.

B A symmetric two-binary variant

In the instance of Theorem 3.1, the improving residuals near zero are all positive, so a positive multiplier halves the required coefficient. This appendix gives a variant in which residuals of both signs occur at cost -1 . Any nonzero multiplier then hurts one of the two sides, the optimal multiplier is zero, and the optimized and zero-multiplier thresholds coincide at 2^{2^d} .

Keep the chain A_d and $\delta_d = 2^{-2^d}$ of (7), the objective $-q$ and the relaxed equality $y = 0$. Add a second binary variable $b \in \{0, 1\}$ and replace the switching inequality in (8) by the pair

$$y \geq a_d - (1 - q) - 2(1 - b), \quad y \leq -a_d + (1 - q) + 2b. \quad (57)$$

The native set is thus $\{(a, y, q, b) : a \in A_d, -1 \leq y \leq 1, q, b \in \{0, 1\}, (57)\}$. At $q = 1$, the choice $b = 1$ forces $y \geq a_d$ and the choice $b = 0$ forces $y \leq -a_d$, so the improving slices lie on both sides of $y = 0$. We write $L_{d,\rho}^{\text{sym}}(\lambda)$ and $D_{d,\rho}^{\text{sym}}$ for the penalized and dual values with the absolute-value penalty.

Proposition B.1. *Let $d \geq 1$. The variant has optimal value 0. For every $\rho \geq 0$ and $\lambda \in \mathbb{R}$, with $c = \rho - |\lambda|$,*

$$L_{d,\rho}^{\text{sym}}(\lambda) = \min\{0, -1 + c\delta_d, -1 + c\}, \quad D_{d,\rho}^{\text{sym}} = \min\{0, -1 + \rho\delta_d\}, \quad (58)$$

and $\lambda = 0$ attains the dual value. Consequently:

- (a) *the least penalty and the zero-multiplier threshold both equal $\delta_d^{-1} = 2^{2^d}$, which has $2^d + 1$ bits, and at $\rho = \delta_d^{-1}$ the multiplier $\lambda = 0$ is the only one at which ρ is exact;*
- (b) *for $0 \leq \epsilon < 1$, the gap $0 - D_{d,\rho}^{\text{sym}}$ is at most ϵ exactly when $\rho \geq (1 - \epsilon)\delta_d^{-1}$;*
- (c) *every $\rho > \delta_d^{-1}$ is minimizer-exact at multiplier zero;*
- (d) *every integer slice (q, b) contains a point at which every native inequality in the continuous variables holds with slack at least $1/8$; for $q = 0$ this point also satisfies $y = 0$.*

Proof. As for Theorem 3.1, we first compute the projection onto (q, y) , on which the objective and the residual depend, and then minimize over it.

Projection. Fix $a \in A_d$. Using $-1 \leq y \leq 1$ and $\delta_d \leq a_d \leq 1$, the four slices give the following y -intervals: $[a_d, 1]$ for $(q, b) = (1, 1)$, since the upper bound $2 - a_d$ is at least one; $[-1, -a_d]$ for $(1, 0)$,

since the lower bound $a_d - 2$ is at most -1 ; $[a_d - 1, 1]$ for $(0, 1)$; and $[-1, 1 - a_d]$ for $(0, 0)$. At $q = 1$ every native point therefore has $|y| \geq a_d \geq \delta_d$. At $a = \bar{a}$ the two intervals at $q = 1$ are $[\delta_d, 1]$ and $[-1, -\delta_d]$, and the two intervals at $q = 0$ are $[\delta_d - 1, 1]$ and $[-1, 1 - \delta_d]$, which cover $[-1, 1]$. Hence the projection is

$$(\{0\} \times [-1, 1]) \cup (\{1\} \times ([-1, -\delta_d] \cup [\delta_d, 1])).$$

Since $\delta_d > 0$, the relaxed equality $y = 0$ forces $q = 0$, and the optimal value is 0.

Penalized value. For a point with residual y , the term $\lambda y + \rho|y|$ is at least $c|y|$, with equality for one sign of y . Both signs occur on each part of the projection, and each part is symmetric in y , so the minimum of $\lambda y + \rho|y|$ over a part equals the minimum of cs over the attainable magnitudes $s = |y|$. At $q = 0$ these are $s \in [0, 1]$, giving $\min\{0, c\}$. At $q = 1$ they are $s \in [\delta_d, 1]$, giving $-1 + \min\{c\delta_d, c\}$. The term c is larger than $-1 + c$, so it can be dropped, which proves the first formula in (58). This expression is nondecreasing in c , and $c \leq \rho$ with equality only at $\lambda = 0$. Hence the dual value is attained at $\lambda = 0$ and equals $\min\{0, -1 + \rho\delta_d, -1 + \rho\}$. Since $\delta_d \leq 1$, the last term can be dropped, which proves the second formula.

Parts (a)–(c). By (58), $D_{d,\rho}^{\text{sym}} = 0$ exactly when $\rho \geq \delta_d^{-1}$, and the same holds for $L_{d,\rho}^{\text{sym}}(0)$. At $\rho = \delta_d^{-1}$ and $\lambda \neq 0$ we have $c < \delta_d^{-1}$, so $-1 + c\delta_d < 0$. The integer 2^{2^d} has $2^d + 1$ bits. For (b), the gap is $\max\{0, 1 - \rho\delta_d\}$, which is at most $\epsilon \geq 0$ exactly when $\rho \geq (1 - \epsilon)\delta_d^{-1}$. For (c), take $\lambda = 0$ and $\rho > \delta_d^{-1}$. A native point with $q = 1$ has penalized value $-1 + \rho|y| \geq -1 + \rho\delta_d > 0$, and a native point with $q = 0$ and $y \neq 0$ has penalized value $\rho|y| > 0$. So the minimizers are exactly the feasible points.

Part (d). Set $a_i = 3/8$ for all i . As in the proof of Theorem 3.1, the chain and its bounds have slack at least $1/8$. On the slices with $q = 0$ take $y = 0$: the inequalities in (57) have slacks $21/8$ and $5/8$ for $b = 0$, and $5/8$ and $21/8$ for $b = 1$. On the slices with $q = 1$ take $y = (2b - 1)/2$: for $b = 1$ the slacks are $1/8$ and $9/8$, and for $b = 0$ they are $9/8$ and $1/8$. The bounds on y have slack at least $1/2$. The smallest slack is $1/8$. $\square$

At $\rho = \delta_d^{-1}$ and $\lambda = 0$, the points $(q, y) = (1, \pm\delta_d)$ have penalized value 0, so this coefficient is exact but not minimizer-exact. The variant has two binary variables, $d + 1$ continuous variables, a linear objective and the same chain of $d - 1$ convex quadratic inequalities as the instance of Theorem 3.1. The fractional-power augmentation of Section 3.4 behaves as before on this variant, because $|y| \geq \delta_d$ on its slices with $q = 1$: the coefficient two is exact, and every larger coefficient is minimizer-exact at multiplier zero.